

# Green development coalition: The new dynamics of rural environmental governance in the Jiangsu-Zhejiang border area of China

Jie Yu<sup>a</sup>, Chao Ye<sup>b,\*</sup>, Liang Zhuang<sup>c</sup>

<sup>a</sup> School of Geographic Sciences, East China Normal University, Shanghai, 200241, China

<sup>b</sup> School of International Relations and Public Affairs, Fudan University, Shanghai, 200433, China

<sup>c</sup> School of Social Development, East China Normal University, Shanghai, 200241, China

## ARTICLE INFO

### Keywords:

Green development coalition  
Cross-border governance  
Village  
Joint river chief system  
Scale

## ABSTRACT

Transboundary environmental governance is fraught with contradictions and tensions. China's Yangtze River Delta is transitioning from economic dominance to green development, and it will provide an important perspective for examining the interactions between economy-environment, central-local, and power-right. In contrast to the growth coalition and green growth coalition—both centered on capital accumulation—this study introduces the concept of the green development coalition, which emphasizes ecological restoration while sustaining local economic growth. Focusing on the Qingxi River, located at the border between Xingqiao Village in Jiangsu Province and Xiyan Village in Zhejiang Province, as a case study, this research examines the environmental governance transformation of Shengze Town and Wangjiangjing Town. Following a history of cross-boundary pollution disputes and independent pollution control efforts, villagers, village committees, and local governments formed a green development coalition, bound together by a “joint river chief system.” The overlapping of multiple national strategies, cross-regional collaboration, social oversight, and grassroots experimental cooperation across boundaries jointly propelled the transition from a growth coalition to a green development coalition. This case offers valuable insights into the innovation of governance models for cross-boundary river management.

## 1. Introduction

There is an inherent tension between economic growth and environmental protection. Since the reform and opening-up, China has experienced remarkable economic growth. Taking economic development as the central task undermines the sustainability of the natural environment (Wu et al., 2014). The regionally decentralized authoritarian system, which combines political centralization with economic regional decentralization, grants local governments a certain degree of autonomy (Xu, 2011). The effectiveness of environmental governance depends on collaboration and mutual compromise between central and local authorities (Qi and Zhang, 2014; Zhou, 2007). Recent literature on China's environmental governance attributes environmental problems to local governments' weak implementation (Zhang, 2017). However, the roles, interests, and interactions between the central and local governments in shaping environmental governance outcomes in China remain unresearched (Kostka and Nahm, 2017). The spatial externalities of pollution—such as air and river pollution—can lead to

“beggar-thy-neighbor” environmental governance between neighboring regions (Chen et al., 2022), adding new complexities to China's severe environmental challenges.

Since 2012, China's central government has prioritized ecological protection, with ecological civilization serving as the ideological framework for environmental policy, law, and education (Xue et al., 2023; Hansen et al., 2018). This marks a shift from “development” to “civilization” in China (Zhang et al., 2023b; Gu et al., 2020). The previously GDP-focused promotion incentive has incorporated environmental targets into the cadre responsibility and assessment system (Kostka, 2016). Local officials must balance economic growth with environmental protection (Yu et al., 2023; Tu et al., 2024). Ecological practices have been continuously emerging. For instance, a river chief system—composed of river chiefs and managers at all administrative levels nationwide—was established to integrate the enforcement capacities of governments and agencies in water resource management (Liu and Richards, 2019). This institutional arrangement has demonstrated significant effectiveness in improving water quality (Wang and

\* Corresponding author.

E-mail address: [yeover@163.com](mailto:yeover@163.com) (C. Ye).

<https://doi.org/10.1016/j.jrurstud.2026.104208>

Received 4 April 2025; Received in revised form 21 April 2026; Accepted 3 May 2026

Available online 7 May 2026

0743-0167/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Chen, 2020). However, as Chien and Hong (2018) point out, the operational model of the river chief system is primarily suited to the management of rivers that traverse prefectural- and county-level administrative boundaries within a single province. Nevertheless, many water pollution incidents cross provincial boundaries, thereby necessitating more innovative measures to extend the applicability of this system.

It is common for local governments and businesses to form alliances to achieve growth through land development, resource consumption, and environmental degradation (Ye et al., 2022). Cross-boundary pollution emerges as a byproduct of this growth-oriented logic. However, under the framework of regional integration, cross-boundary pollution—once neglected—has become a major barrier to ecological integration, and the importance of cross-boundary environmental governance has gradually increased (Hou et al., 2021). As the central government of China has incorporated environmental metrics into performance evaluations and deployed a series of strategic regional policies, environmental governance, particularly transboundary issues, has become an unavoidable concern for local governments, extending beyond their growth agendas (Han and Sheng, 2024). Under the leadership of the central government, horizontal ecological compensation between local governments has effectively integrated bilateral rights, responsibilities, rewards, and punishments, thus merging watershed management with administrative territorial management (Yu et al., 2024). Regional carbon trading markets have provided institutional tools for simultaneously achieving environmental sustainability and sustained economic growth. In addition, self-governance, through context-specific rules, can cope with asymmetries in the commons and offers insights for regional governance (Ostrom and Gardner, 1993). Consequently, how to coordinate intergovernmental relations to achieve effective cross-boundary environmental governance has become a critical issue.

The concept of coalition in regional governance research is continuously expanding its scope of application, with cross-boundary regions, rural areas, and ecological issues emerging as new research frontiers. Classic discourses such as the “growth coalition” have been particularly influential in revealing the political-economic foundations of local development. The growth machine theory posits that local governments and elites form alliances to facilitate land development and urban expansion in pursuit of economic growth (Molotch, 1976). This perspective has been utilized by researchers to explain China's local economic expansion (Zhang, 2014). With growing concerns over global environmental degradation, environmental protection and sustainable development have been increasingly integrated into urban development strategies. Recent studies have explored the “greening” of the urban growth machine by incorporating environmental objectives (Scanu et al., 2021; Shao et al., 2025; Zhao, 2024), giving rise to concepts such as the “green growth coalition.” These coalitions embed green discourse into urban governance, enhancing land value through the creation of ecological landscapes and “green labeling,” with the aim of facilitating urban growth and capital accumulation (Jocoy, 2018). In contrast to market-oriented neoliberal governance, China exhibits a model of state-centered governance characterized by “state entrepreneurialism.” Scholars have used “authoritarian environmentalism” to describe China's environmental governance, which is characterized by campaign-style mobilization, target-setting, and control mechanisms in achieving central-initiated environmental goals (Feng et al., 2024). Historically, local governments and market actors have collaborated in the pursuit of economic growth. Under the environmental agenda initiated by the central government, more comprehensive objectives related to socio-ecological sustainability have been incorporated into local governance frameworks (Zhang et al., 2022b). Issues that were previously marginalized, such as transboundary and rural environmental problems, have now been brought within the scope of governance. This challenges the traditional, growth-oriented “urban growth machine.” Applying growth coalition and green growth coalition

theories to China's rural border regions—analyzing their applicability in past practices and exploring their potential under the sustainability paradigm—can enrich and expand the theoretical boundaries of the growth machine itself.

China's rapid industrialization has been accompanied by significant environmental, health, and social challenges (Green and Stern, 2017). Compared with urban areas, rural China exhibits a wider distribution of industrial enterprises, higher intensities of air and water pollution, and elevated environmental risks (Long et al., 2022). However, the direct participation of these market forces—which act as the primary polluters—remains extremely limited in rural environmental governance decision-making and public affairs. Driven by incentives for investment attraction and industrial expansion, some rural areas maintain relatively lax oversight of corporate environmental behavior, often subordinating environmental governance to economic objectives in practice (Zhang et al., 2023a). Furthermore, deficiencies in grassroots governance capacity and variations in regulatory enforcement constrain the capacity of administrative authorities to effectively regulate corporate behavior (He et al., 2014). Additionally, limited awareness of environmental rights and weak social supervision mechanisms in rural society further undermine external constraints on corporate behavior (Yu and Ye, 2025). Under the combined influence of these factors, enterprises lack the institutional incentives and social pressures to proactively engage in environmental governance. Consequently, they primarily adopt a stance of passive regulatory compliance, leading to their limited participation in rural public governance processes. For a long time, a disproportionate volume of resources, including capital, institutional capacity, and environmental regulations, has been allocated to urban areas rather than rural ones (Wang et al., 2008). Rural societies maintain a high degree of direct livelihood dependence on ecosystems, including rivers and land (Berrouet et al., 2020). Environmental degradation can translate rapidly into risks for social stability and sustainable development. In Western contexts, rural governance initiatives are primarily initiated and led by local communities, social organizations, or industry associations through bottom-up approaches, as exemplified by the EU's LEADER programme (Johansson and Holmquist, 2024; Scott, 2004). In contrast, rural public affairs in China have long been dominated by the state. Local governments bear primary fiscal responsibility for the provision of rural public goods. Furthermore, the fact that Chinese rural collective land cannot directly enter the real estate market weakens local governments' capacity to drive growth through land marketization (Jiang and Lin, 2021). Consequently, rural governance processes rely more heavily on negotiation and cooperation between local governments and village collectives (Shen and Shen, 2018). The resulting coalition structures focus predominantly on public affairs consultation, resource integration, and institutional embedding. Moreover, rural development and governance are increasingly influenced by urban and broader regional dynamics (Wang et al., 2023, 2025; Zhang et al., 2022a). In the Yangtze River Delta, for instance, industrial systems, migratory flows, and ecosystems are highly coupled. Environmental issues in this region exhibit significant transboundary characteristics, externalities, and complexity. This is particularly evident in villages located at administrative boundaries, where environmental changes reflect the ongoing transformation of China's environmental governance.

Traditional “growth coalition” frameworks, as well as “green growth coalition” extended from urban contexts, are no longer sufficient to capture the dual emphasis on public interests and development in Chinese rural governance, or to explain the hybridity and cross-boundary characteristics observed in cross-boundary areas. Environmental governance in cross-boundary regions simultaneously involves spatial externalities, cross-actor interest conflicts, and multi-level mechanisms of power transmission. In these areas, environmental issues are inherently transboundary, disrupting governance logics traditionally rooted in territorial management. Local governments exercise administrative power within jurisdictional boundaries, yet their actions generate

externalities beyond them; affected residents bear the environmental consequences but lack the institutional capacity to intervene directly. There is a mismatch between power allocation and rights realization. This misalignment is further amplified within the multi-level bureaucratic hierarchy, where grassroots demands must move upward through successive tiers to be redefined and coordinated. Furthermore, the term “cross-boundary areas” in this study specifically denotes subnational administrative border regions, such as provincial or municipal border zones, where fragmented jurisdictional authority intersects with highly integrated socio-economic and ecological systems. Consequently, this study proposes the concept of the “green development coalition.” We use this framework to conceptualize and explain the multi-scalar actor configurations, institutional arrangements, and governance practices observed in rural environmental governance within China's border areas. Specifically, we aim to understand how growth coalitions are formed in rural areas, and to examine how the composition and operational modes of growth coalitions are transformed when an economic expansion-oriented logic of “growth” encounters state-led environmental objectives centered on ecological restoration. The central research question of this study is as follows: in China's cross-boundary rural environmental governance, how are green development coalitions constructed and operated through the interactions among governments, enterprises, and residents?

This study summarizes the governance process of a transboundary river in Jiangsu and Zhejiang Provinces, examines the factors influencing this governance process, and traces the local transition from growth coalition to green development coalition. The study explores the evolution of local economic-environmental relationships within the context of China's rapid urbanization and industrialization. It reveals how regional dynamics have shifted from antagonism to cooperation under the framework of multiple national strategies. It shows a combination of top-down and bottom-up social management systems. These findings provide theoretical and empirical insights into local environmental governance within the broader framework of ecological modernization.

## 2. From the growth coalition to the green development coalition

### 2.1. Growth coalition under rapid urbanization and industrialization

The growth machine thesis is an influential explanation of land-use decisions that contribute to the understanding of urban political processes under rapid urbanization and industrialization (Jocoy, 2018). Harvey Molotch first introduced the concept of the “growth machine” in 1976, emphasizing that the political and economic essence of place is growth. The desire for growth leads local governments, local businessmen, property owners or investors in locally oriented financial institutions, newspaper publishers, leaders of public or quasi-public agencies, and some local elites to form growth coalitions (Molotch, 1976). The growth coalition seeks to transform the city into a “growth machine” by placing the exchange value of land above the use value. Local governments are the most important stakeholders, and the “development of growth” is their primary concern. To resolve the conflict between the use and exchange value of land, “through their control of local government and ideological leadership in matters of land use and production, rentiers and their coalitions mediate between capitalist investors and other social classes.” (Logan and Molotch, 1987). The “machine” metaphor conceptualizes the driving forces of urban development as institutionalized and structural, whereas the notion of “coalition” emphasizes more concrete political-social networks, revealing how actors, through their practices, propel the functioning of the machine (Molotch, 1993; Jonas and Wilson, 1999).

The concept of “growth coalition” provides a tool for explaining the relationship between government operations and business interests for local economic development. China's growth miracle has been accompanied by a dramatic reshaping of the relationships between industry

and agriculture, urban and rural areas, and different regions (Lin, 2002). The institutional reforms initiated in 1978 were aimed primarily at increasing productivity. Socialist marketization and decision-making decentralization sparked the emergence of a new political economy and fostered “local state corporatism” (Oi, 1995). Local governments, enterprises, and communities established shared interest networks and stable capital accumulation mechanisms. Since the 1990s, fiscal decentralization in China has granted local governments increased economic autonomy. Marketization and privatization have accelerated capital-led county-level urbanization. Local governments offer cost-effective land, a favorable business environment, and tax support for industry and commerce. Businesses create jobs and profits, solve local unemployment, and bolster local government fiscal returns (Yu and Ye, 2025). Thus, a close government–enterprise nexus emerged. Local space is organized through economic growth, and the essence of this economic activity is a growth machine.

While the “growth machine” model analysis has been widely deployed to explain urban changes, the accent placed on coalitions internal to a regime often leads to a neglect of the determinate social forces operating beyond a particular urban scale (MacLeod and Goodwin, 1999), including geographic situations, regional variations, and locational patterns (Lin, 2002). It prioritizes growth imperatives in urban politics while paying little attention to the changing social issues of non-growth. In practice, the expansion of growth coalitions inevitably leads to damage to public interests, such as environmental pollution, traffic congestion, and reduction of green spaces (Molotch, 1993). This means that only a small number of groups benefit from local economic growth, with a much larger population passively bearing environmental costs disproportionate to the economic gains. Such disparities have deepened social divisions, fueling the rise of environmentalists as an anti-growth coalition. Subsequent anti-growth movements may in turn disrupt the functioning of the growth machine and challenge its dominance in urban governance.

### 2.2. Green growth coalition in the context of sustainable development

The application of the growth machine to urban governance has promoted population expansion and economic development at the cost of ecological and social sustainability. Environmental challenges such as pollution and resource degradation have influenced urban regimes and policy strategies. The green growth machine as a new economic growth strategy offers a potential solution to the tension between economic expansion and local environmental concerns (While et al., 2004). It builds upon the traditional growth machine framework. Growth advocates incorporate green values and continuously modify growth discourses to fit changing political contexts while persuading environmentalists and other civic groups to join green growth coalitions (Jocoy, 2018). They redevelop the space by building amenities that generate higher economic value and profit accumulation (Sbicca, 2019), charging substantial premiums for land investment, real estate, and so on (Dilworth and Stokes, 2013).

The green growth machine challenges the traditional growth machine's model of capital accumulation by shifting from land development, resource extraction, and real estate construction to sustainable technologies and low-carbon industries. However, the emphasis on economic growth and technological innovation and the neglect of social inequality and poverty have led to the “greenwashing” debate (Munshi and Kurian, 2005). Some developments labeled as “green” or “sustainable” may serve as marketing tools rather than genuine commitments to environmental protection or social sustainability (Checker, 2011). Green space projects that appear to be ideology value-free may subordinate social equity or public sector planning to lucrative real estate development (Anguelovski et al., 2018). Resilient green infrastructure that raises land and real estate values is built for well-heeled newcomers for “higher uses” (Rigolon and Németh, 2018; Lim et al., 2013; Shao et al., 2025). Seemingly laudable greening initiatives might become

“disruptive green landscapes,” leading to green gentrification (Anguelovski et al., 2022; Triguero-Mas et al., 2021). Low-income and vulnerable social groups may be marginalized or even excluded from the circle of benefits from the green transition, leading to displacement (DuPuis and Greenberg, 2019).

The green growth machine is rooted in the market logic of neoliberalism. It prioritizes the interests of the green growth machine without considering the resulting social justice issues like green gentrification, poverty, and ecological burden shifting. This framework overlooks the necessity for fundamental institutional restructuring and fails to account for the structural transformations experienced by China in recent years. This shift transcends simple industrial adjustment; it represents a fundamental evolution in governance, moving from a growth-centric model toward an integrated paradigm that balances economic development, environmental protection, and social equity. Crucially, this requires a recalibration of the state's redistributive role, shifting the strategic focus toward enhancing the ecological well-being of residents in underdeveloped and rural areas. Closely tied to economic development and industrial transformation, the green growth machine emphasizes market mechanisms and partnership-based actions to drive green transitions. However, it underestimates the leading role of public policy and neglects the importance of social mobilization and civic participation. Additionally, since different regions face varying environmental challenges, resource endowments, and developmental stages, the green growth machine framework may not be universally applicable across all contexts.

2.3. Proposal of the green development coalition

The growth machine framework depicts the speculative behavior of local elites and governments who form stable alliances to capture land value appreciation and facilitate economic expansion (MacLeod, 2011). This theory has primarily focused on urban areas, frequently marginalizing the associated environmental impacts (Table 1). While the green growth coalition introduces an ecological dimension, it remains essentially an extension of the “urban growth machine”; its core logic centers on using environmental modification to sustain accumulation and profitability (Lang and Rothenberg, 2017). Both frameworks facilitate the continuous profitability of capital within the urban space. In response to the environmental crisis, China has established environmental targets centered on comprehensive development. Consequently, local governments must navigate a strategic trade-off between local economic growth and national social stability. As a result, previously marginalized transboundary ecological spaces have been integrated into the governance agenda, leading to emerging shifts in the configuration of local coalitions. Building upon a dialogue with the theories of growth coalitions and green growth coalitions, we propose the concept of the “green development coalition.” The green development coalition is an integrated governance network composed of government officials, enterprises, residents, and social organizations—that engages in

transboundary multi-stakeholder collaboration to carry out socio-ecological restoration in marginalized cross-boundary ecological spaces to achieve comprehensive development. “Urban entrepreneurialism” conceptualizes the city as a vehicle for capital accumulation, emphasizing speculative development and inter-urban competition in pursuit of economic growth (Harvey, 1989). By contrast, the public goals of green development coalitions—such as environmental sustainability, social equity, and ecological well-being—play a central role in shaping interactions among actors and guiding collective action. From this perspective, the concept captures a collaborative mechanism with strong public governance attributes, rather than a set of policy instruments or deliberately designed strategies employed by individual actors.

The concept of the green development coalition emphasizes the dual objectives of economic development and ecological restoration. It incorporates the connotations of the “developmental coalition” proposed by Zhang and Gao (2022), highlighting a paradigm shift from the singular logic of economic growth—embedded in traditional urban growth coalitions—toward a collective action framework oriented by public value (Ansell and Gash, 2008; Bulkeley, 2005). Unlike the “spatial fix” for resolving capital over-accumulation (Harvey, 1981, 2001), or the “sustainability fix” which selectively incorporates environmental goals into capital accumulation (While et al., 2004), the green development coalition achieves its “green” objectives through socio-ecological restoration. This involves the modification, management, and restructuring of the natural environment to stabilize the political-economic order in response to ecological and social crises (Ekers and Prudham, 2017; Comby et al., 2019; Zhang et al., 2022a). Its governance scope extends beyond urban environmental management to include rural hinterlands and transboundary spaces. Rural areas have long been marginalized as mere supply zones for urban areas (Ye and Liu, 2020). These regions differ from urban areas in management systems, community structures, and land-use patterns; they possess fragile ecosystems and face higher health risks (Zhao et al., 2019; Yan and Chen, 2024; Long and Liu, 2016). The green development coalition aims to restore rural ecosystems and reconstruct local value. In addressing transboundary issues characterized by high mobility and negative externalities—such as air and river pollution—the green development coalition operates by transcending administrative boundaries. It seeks to establish collaborative cross-regional systems that shift from localized patches of restoration toward holistic regional ecological governance.

While the growth coalition and green growth coalition focus on coalitions between local entities and ignore the top-down and bottom-up negotiation processes in China, the green development coalition draws on multi-level governance theory to emphasize a coordination mechanism nested within diverse governance scales (Hooghe and Marks, 2003; Bulkeley and Betsill, 2005). Rather than being a policy instrument dominated by a single tier of authority, it functions as a catalyst that activates both state and non-state actors across different jurisdictions and levels. This enables a reconfiguration of resources, power, and knowledge within the transboundary governance process (Brenner, 2004). Such governance dynamics transcend simple hierarchical delegation or bottom-up mobilization; instead, through continuous negotiation and institutional nesting, they reshape the vertical interactions between governmental tiers and the relationships between state and social actors. The stability of the coalition hinges on the consensus and compromise achieved among stakeholders regarding social, economic, and environmental sustainability. Taking the “joint river chief system” as an example, this transboundary collaboration achieves not only vertical synergy across administrative hierarchies but also the downscaling of ecological governance to the micro-level of the village. In doing so, the coalition reconstructs local public value and pioneers a new mode of territorial governance—one characterized by state leadership, multi-stakeholder participation, and shared benefits.

Participants in the green development coalition include governments, enterprises, residents, the media, and civil society organizations.

Table 1  
Key dimensions of Growth, Green Growth, and Green Development Coalitions.

|                       | Growth Coalition     | Green Growth Coalition     | Green Development Coalition                   |
|-----------------------|----------------------|----------------------------|-----------------------------------------------|
| Focus                 | Urban land           | Urban green space          | Marginalized cross-boundary ecological space  |
| Value orientation     | Exchange value first | Green as a tool for growth | Comprehensive development                     |
| Core driving force    | Land appreciation    | Profit via green discourse | Transboundary multi-stakeholder collaboration |
| Stance on environment | Ignored              | Opportunity                | Core component of development                 |
| Source                | Molotch (1976)       | Dilworth and Stokes (2013) | Proposed in this study                        |



Throughout its formation and evolution, these actors exhibit differentiated motivations, strategic logics, and behavioral dynamics. In the initial stage, local governments, constrained by growth-oriented performance evaluation systems, tend to relax environmental regulations to attract mobile capital. Enterprises, driven by cost-minimization imperatives, externalize production-related pollution to other regions, while residents in neighboring jurisdictions bear the resulting environmental and health consequences. This misalignment of interests renders inter-local relations a “zero-sum game,” with conflicts manifesting as environmental resistance by affected groups and responsibility shifting by polluting actors. Affected residents initially articulate their claims through institutionalized channels, such as petitions and complaints. However, when these channels prove ineffective, grievances may escalate into media exposure and collective action. In other words, when rights infringements caused by environmental externalities cannot be resolved through horizontal negotiation within a given administrative scale, affected actors may adopt strategic actions to elevate geographically bounded conflicts onto higher-level political agendas. When local conflicts transcend jurisdictional boundaries, “scale-up” occurs, triggering vertical intervention by higher-level governments. Through top-down mechanisms—such as environmental target-setting, performance evaluation, regulatory enforcement, and cross-regional coordination—upper-level authorities recalibrate the incentive structures faced by local actors. This process encourages local governments to shift from competition-driven, growth-oriented strategies toward more cooperative approaches, particularly in addressing transboundary environmental problems that cannot be resolved unilaterally. In this sense, scale jumping functions as both a mechanism of pressure transmission and institutional activation, linking grassroots social demands with higher-level administrative intervention. It is through this dynamic process of conflict escalation and institutional response that the green development coalition emerges. Local governments establish cross-regional collaboration mechanisms—such as joint law enforcement, ecological compensation, and planning coordination—to horizontally align interests across administrative units while vertically integrating policy resources and governance instruments. Enterprises, subject to both policy incentives (e.g., subsidies and tax preferences) and regulatory pressures (e.g., emission standards and penalty regimes), gradually transition from “passive compliance” to “proactive adaptation,” and may even participate in the formulation of green production standards. Meanwhile, residents and social organizations become embedded in governance processes through institutionalized channels—such as public participation and information disclosure—thereby linking internal protest with external oversight. Ultimately, through sustained interaction and adaptive adjustment, the coalition evolves into a relatively stable, self-organizing governance network. Enterprises gain new competitive advantages through green transformation, local governments enhance overall regional performance through collaborative governance, and residents benefit directly from environmental improvements, resulting in a mutually beneficial configuration of shared gains among multiple actors.

The shift from growth coalitions to green development coalitions in China reflects the combined influence of environmental crises, policy imperatives, evolving local awareness, and the rise of civic environmentalism, embodying a reshaping of state-local-society relations. The green development coalition is not perceived as a spontaneous coalition formed through horizontal negotiations among equal participants. Instead, it represents a government-led structure that incorporates multiple stakeholders, where state authority, regulatory instruments, and performance-based incentives realign the roles, interests, and interactions of local governments, enterprises, and residents toward ecological restoration goals. Specifically, China's central government has elevated environmental governance into a national strategic priority through top-down target setting, mandatory regulatory mechanisms, and performance accountability systems. In turn, local governments provide support for ecologically fragile areas by investing in green

infrastructure, offering financial subsidies, and promoting green industries. Meanwhile, residents constitute an integral component of this coalition; they act not merely as passive recipients of environmental outcomes but as agents with explicit environmental demands related to health, livelihoods, and daily well-being. Civil society complements, and sometimes corrects, formal governance processes through collective petitions, complaints, and protests (Guttman et al., 2018).

Driven by the combined pressures of severe environmental challenges, top-level state strategies for regional cooperation, and diverse societal demands for environmental protection, inter-local relationships have gradually shifted from competition toward cooperation—particularly in cross-boundary environmental governance. It is the outcome of a dynamic interplay between “power” and “rights.” Here, “power” refers to the administrative intervention, resource allocation, and policy enforcement exercised by central and local governments within a bureaucratic system, whereas “rights” denote the legally grounded entitlements and interests of citizens, including environmental rights. The tension between power and rights stems from the mismatch between the territorially bounded operation of administrative authority and the inherently cross-boundary nature of environmental problems. Under GDP-oriented incentive structures, local governments tend to prioritize economic growth and may exercise administrative power by relaxing environmental regulations or externalizing pollution costs. However, such environmental costs often transcend administrative boundaries, disproportionately affecting residents in downstream or adjacent areas. As a result, a structural condition emerges in which those who bear environmental risks are spatially separated from those who exercise regulatory authority, thereby imposing institutional constraints on the realization of citizens' environmental rights. In this context, citizens articulate their rights claims through a combination of formal and informal channels. Affected communities initially rely on formal mechanisms—such as petitions, complaints, and administrative appeals—to seek redress. When these channels prove ineffective due to fragmented jurisdiction or regulatory insufficiencies, rights claims may escalate into more visible and collective forms of action, including protests, media exposure, and cross-regional disputes. This process should not be understood merely as confrontation; rather, it constitutes a mechanism through which grassroots pressures are embedded into a hierarchical governance system. On the one hand, governments in affected areas engage in negotiations with authorities in pollution-source regions in response to local demands; on the other hand, they seek support from higher-level governments, thereby mobilizing superior authority and resources. In response to regional environmental risks, upper-level governments intervene, coordinate across jurisdictions, and recalibrate fragmented governance arrangements. In this sense, administrative power is not only a source of environmental governance problems but also a crucial institutional guarantee for the realization of citizens' rights. Under the dual pressures of sustained grassroots mobilization and top-down accountability mechanisms, existing configurations of power are restructured, prompting local governments to shift from a singular growth-oriented approach toward governance models that place greater emphasis on environmental responsiveness and regional coordination. It is precisely through this process that the tension between power and rights is transformed into a driving force for institutional adjustment and governance innovation, ultimately giving rise to a cross-regional green development coalition involving multiple actors, including local governments, village-level organizations, residents, and the media.

In contrast to the “fragmented authoritarianism” in environmental governance and the tendency of existing research to attribute the “tragedy of the commons” to non-cooperative games between local governments, the green development coalition delineates the process by which local governments transform external pressures into an internal will for cooperation. It elucidates the construction of an organized, goal-oriented, and cross-boundary networked structure, providing a novel perspective for understanding the dynamics and institutions of cross-

regional collaborative governance. It demonstrates how grand national objectives are effectively translated into local action plans through the participation and benefit-sharing of multiple regions, tiers, and groups, representing an organic integration of state will and local wisdom. Unlike traditional studies that focus on intergovernmental coordination, the green development coalition emphasizes the agency of grassroots actors—including village cadres and residents—within the environmental governance structure. It provides a bottom-up analytical perspective and offers empirical evidence for understanding the governance capacity of Chinese grassroots society and its value in regional collaborative governance. Transcending the conventional “state-society” dualistic framework, the insights offered here are applicable not only to the Chinese context but also serve as a reference for green transition pathways in other Global South countries. However, the sustained existence of the green development coalition beyond cyclical policy campaigns depends on whether ecological goals can be normalized within local governance practices, cross-jurisdictional coordination mechanisms, and economic incentive structures, rather than relying solely on cyclical administrative mobilization. Although ecological restoration yields collective, long-term environmental benefits, the governance costs of enforcement and coordination are borne unevenly by various parties. Local governments face increasing administrative and fiscal burdens due to cross-regional coordination, monitoring, and performance evaluation. The motivation for non-state actors to maintain participation depends not only on normative commitments to environmental improvement but also on whether compensation mechanisms, fiscal transfers, or long-term economic returns can offset short-term losses.

3. Study area and data collection

3.1. Study area description

Xiyan Village of Zhejiang Province is located on the western side of Wangjiangjing Town (Fig. 1). The village covers an area of 2.88 square kilometers, with 13 village groups and 456 households. Xingqiao Village of Jiangsu Province is situated at the southern end of Shengze Town. The total area of the village is 5.6 square kilometers, with 22 village groups and 744 households. Wangjiangjing Town, with an administrative area of 127.3 square kilometers, comprises 33 administrative villages and 9 residential communities, sustaining a permanent population of 139,000.

In 2022, its gross regional product (GRP) reached 10.6 billion yuan, accompanied by 19.68 billion yuan in industrial output from enterprises above designated size. In comparison, Shengze Town spans a larger area of 150 square kilometers, encompassing 30 administrative villages and 8 residential communities, yet supports a significantly higher permanent population of 400,000. The town achieved a GRP of 46.428 billion yuan in 2022, with industrial enterprises above the designated size generating 61.448 billion yuan in output value, demonstrating an economic scale 3.4 times that of Wangjiangjing Town. To clarify the administrative hierarchy of the study areas, Table 2 delineates the four-tier administrative divisions of Xiyan Village and Xingqiao Village.

The Qingxi River (referred to as the Qingxi River in Jiangsu Province and Maxigang in Zhejiang Province) serves as the boundary river between Jiangsu and Zhejiang provinces, connecting upstream Shengze Town with downstream Wangjiangjing Town. The Xiaotianwei River, an inland branch of the Qingxi River, also functions as the boundary between Xiyan Village and Xingqiao Village. In the past, the Xiaotianwei River suffered from poor water quality, appearing turbid and emitting an offensive odor. Local villagers built an earthen dam to facilitate crossing, which disrupted water flow and led to sediment accumulation. In 2023, joint ecological restoration efforts by Wangjiangjing Town and Shengze Town significantly improved the river's environmental conditions.

3.2. Data collection

We conducted a field survey from February 2024 to January 2025. The research methods were non-participant observation, semi-structured interviews, and focus groups. Semi-structured interviews were conducted with a total of thirty persons (Table 3). The interviewees' ages and backgrounds were stratified, including government

Table 2  
Administrative divisions of Xiyan Village and Xingqiao Village.

| Village          | Town          | District/County-level City | Prefecture-level City | Province |
|------------------|---------------|----------------------------|-----------------------|----------|
| Xiyan Village    | Wangjiangjing | Xiuzhou District           | Jiaxing               | Zhejiang |
| Xingqiao Village | Shengze       | Wujiang District           | Suzhou                | Jiangsu  |



Fig. 1. Location of Xingqiao village and Xiyan village.  
Sources: The national and provincial maps are from the Standard Map Service of the Ministry of Natural Resources of the People's Republic of China (Approval No. GS (2023) 2767). The base map has not been modified.

**Table 3**  
Information of the interviewees.

| Town          | Location of the interview     | Interviewee       | Code | Gender | Number |
|---------------|-------------------------------|-------------------|------|--------|--------|
| Wangjiangjing | Xiyan Village                 | Villagers         | WV1- | Female | 5      |
|               |                               |                   | WV8  | Male   | 3      |
|               | Xiyan Village                 | Village Committee | WC1- | Female | 2      |
|               |                               |                   | WC3  | Male   | 1      |
|               | Wangjiangjing Town Government | Officers          | WG1- | Female | 2      |
|               |                               |                   | WG3  | Male   | 1      |
| Shengze       | Xingqiao Village              | Villagers         | SV1- | Female | 4      |
|               |                               |                   | SV6  | Male   | 2      |
|               |                               | Factory owners    | SO1- | Male   | 2      |
|               |                               |                   | SO2  |        |        |
|               | Xingqiao Village              | Migrants          | SW1- | Male   | 2      |
|               |                               |                   | SW2  |        |        |
|               |                               | Village Committee | SC1- | Female | 2      |
|               |                               |                   | SC3  | Male   | 1      |
|               | Shengze Town Government       | Officers          | SG1- | Female | 1      |
|               |                               |                   | SG3  | Male   | 2      |

officials, village cadres, factory owners, and villagers, ensuring a diversity of perspectives. A purposive sampling strategy was used to select interviewees, focusing on village cadres and local officials who were directly involved in transboundary governance events (such as the joint river chief system and the Xiaotianwei project), as they could provide the most relevant first-hand information. Two-thirds of the interviewees had lived or worked in Shengze Town or Wangjiangjing Town since the 2000s, enabling them to witness or participate in the region's development. The interview protocol was developed based on preliminary fieldwork and literature review. Core questions revolved around environmental changes, governance practices, industrial production, cross-boundary cooperation, and villagers' perceptions. The questions were designed to be both open-ended and guiding, which allowed interviewees to express themselves freely while still covering the research topics. All face-to-face interviews lasted more than 30 minutes. During the interviews, the researcher provided interpretive feedback to the respondents for confirmation and concluded the interviews once information saturation was reached. To minimize the limitations and biases of single-source data, the study also collected secondary materials, including statistical yearbooks, news reports, official government website data, and materials from the "Joint River Chief System Exhibition Hall." Data analysis employed thematic coding, with two researchers independently conducting initial coding and then cross-checking the results. Any discrepancies were resolved through discussion. Triangulation with participatory observation and documentary evidence was used to enhance the reliability of the findings. For validity, the researchers continuously reflected on their own process and invited some interviewees to review the study's conclusions.

This study attempts to explain the tension between economic growth and environmental protection in China's urbanization and industrialization process, analyze the conflicts between grassroots groups and growth coalitions, and reveal the local transition from a growth coalition to a green development coalition. We conceptualize the Qingxi River as a social-ecological system, where ecological processes, administrative boundaries, and socioeconomic activities are closely intertwined. Its ecological degradation and restoration are not merely physical phenomena but are profoundly influenced by institutional arrangements, cross-jurisdictional relations, and local development strategies. We take the governance transformation of the Qingxi River as a typical case to analyze shifts in China's environmental governance paradigm for three reasons. First, since the reform and opening-up, Shengze Town in Jiangsu Province and Wangjiangjing Town in Zhejiang Province have actively responded to the central government's call to promote industrialization. They are representative regions of China's industrialization process. The resulting pollution issues are highly typical and reflect broader environmental challenges across the country. Second, the

Qingxi River serves as the boundary between Jiangsu and Zhejiang, both pollution and governance inevitably involve two administrative regions. The governance process is characterized by the coordination of multiple regional governments, various governmental departments, and different levels of administration. The experience provides a valuable reference for the management of transboundary rivers in other regions, such as the Pearl River Delta and the Yellow River Basin. Finally, grassroots collective action has played a crucial role in improving the water quality of the Qingxi River, highlighting the supplementary role of non-state actors and the evolving perception of environmental governance in rural areas. It provides new insights into the collaborative participation of multiple stakeholders in environmental governance.

#### 4. Environmental pollution of the Qingxi River and the rural growth coalition

##### 4.1. Frequent transboundary pollution under rapid industrialization

Shengze Town in Jiangsu Province and Wangjiangjing Town in Zhejiang Province have a long history of textile industry. Residents historically engaged in silk weaving and cotton spinning handicraft production. Since China's reform and opening-up in 1978, the two localities vigorously developed the textile industry. Shengze Town formed a comprehensive industrial system from research and design, weaving and printing, and dyeing to education services in the 1980s. At that time, "every village runs its own factory" was a common sight. In 1989, there were 146 enterprises in Shengze Town, of which 86.3% were in the printing and dyeing textile industry, and more than 90% of the employees were directly or indirectly related to the silk industry. From 1989 to 1999, the number of enterprises surged to 419, with total industrial output value rising from 730 million yuan to 4.506 billion yuan. Since 2000, Shengze Town has attracted many industrial enterprises. It has gradually developed into a silk chemical fiber and textile industry cluster. A 60-square-kilometer economic development zone has been established, and two Fortune Global 500 companies have come out of the town.

Wangjiangjing Town has a total water area of 30.8 square kilometers, with a water coverage rate of 24.2%. It is a major freshwater fishery production area in Zhejiang Province, boasting an aquaculture area of over 1900 ha. The town is especially known for its black carp (*Mylopharyngodon piceus*) farming, ranking among the top in both scale and production at the provincial and national levels. In 1983, farmers from Tianle Commune took the lead in purchasing an iron-wood loom (powered) and established household workshops, marking the beginning of the rural household textile industry. This traditional household textile industry rapidly expanded in the following years. During the 1990s, the government made big moves to support the growth of individual and private economies, fueling the rapid development of the town's household textile industry. By 2000, Wangjiangjing Town had a total of 6463 individual and private industrial enterprises, including 50 private enterprises and 6413 individual enterprises. Fish and textile are the most notable labels of the town.

Rapid industrialization has reshaped the spatial structure of both towns, intensifying the economic and social imbalance, with water pollution being a key consequence. The Qingxi River is a waterway with a width of 30 m and a depth of 2 m. It flows from west to east through Shengze Town in Jiangsu Province and flows into Wangjiangjing Town in Zhejiang Province, closely connecting the two towns. The textile dyeing and printing industry is characterized by high pollution and high emissions, generating smoke, sulfur dioxide, and significant volumes of intractable industrial wastewater. While Shengze Town has aggressively expanded its textile dyeing and printing sector, its environmental infrastructure, including sewage treatment facilities and pipeline networks, failed to keep pace. As a result, untreated or inadequately treated dyeing wastewater has been directly discharged into rivers flowing toward Zhejiang, severely polluting the northern waters of Wangjiangjing

Town. In the late 1980s, the water grade at the border of Jiangsu Province and Zhejiang Province was categorized as Grade III, but it deteriorated sharply year by year in the 1990s. Since 1992, the water quality has been worse than Grade IV, and since 1996, it has fallen below Grade V, indicating extreme pollution levels, rendering the water unsafe for human contact. This has severely affected fishermen's livelihoods and threatened the town's traditional freshwater aquaculture industry.

#### 4.2. Pollution control in two places after transboundary conflicts

Pollution has heightened tensions between Wangjiangjing Town and Shengze Town. Residents of Wangjiangjing Town, represented primarily by fishing communities, once lodged complaints and petitions with the Wangjiangjing Town Government, the Xiuzhou District Government, and the Jiaxing Municipal Government. However, due to the territorial jurisdiction system, the Zhejiang side had no authority over the factories and enterprises located within Shengze Town. In 1995, a large-scale fish die-off occurred in Wangjiangjing Town. In a fit of anger, the fishermen piled a large number of dead fish at the Shengze Town government building. This action, however, had little effect, and the pollution continued. By 2001, water pollution incidents had escalated. *"By 2000, the water in our river had turned a murky green, almost black. It was all wastewater from dyeing factories, discharged directly."* (Interviewee WC1) *"When the wastewater came, all the fish and pearl mussels died. In other words, they (the dyeing factories) destroyed the fishermen's livelihoods."* (Interviewee WV1)

During the periods of late April to early May, late August, and late September to the end of October in 2001, massive amounts of wastewater were discharged into Wangjiangjing on three separate occasions. These events contaminated 2000 ha of water bodies and caused an estimated economic loss of 56 million yuan. On the night of November 21, 2001, three hundred villagers around Wangjiangjing Town raised 1 million yuan, used eight bulldozers and tens of thousands of sacks, and sank 28 cement boats to block the Qingxi River. The villagers held up banners that read, "Return our clear rivers! Give back our fertile land!" and "A few bosses got rich, but millions suffered." They spontaneously organized a team of 100 people to protect the dam, guarding it day and night. This event is known as the "Grassroots Midnight Action".

Residents of Wangjiangjing Town protested through more intense forms of collective action, which broke through hierarchical constraints, escalated the issue to a higher level, and shifted the power dynamics. This process reflects the inherent tension between the realization of citizens' environmental rights and the operation of local administrative authority. Under growth-oriented incentive structures, the Shengze Town government exercised its administrative power over industrial development and environmental regulation, while downstream communities bore disproportionate ecological and economic costs, giving rise to rights-based environmental claims. This misalignment produced a structural condition in which those affected by pollution lacked direct regulatory authority, whereas those possessing such authority did not fully internalize its consequences. The resulting tension became increasingly visible through a process of escalating claims-making. Initially, residents of Wangjiangjing Town relied on institutionalized channels—such as submitting petitions and complaints to township, district, and municipal governments—to seek redress. Notably, as the territorially responsible authority, the Wangjiangjing Town government's administrative logic at this stage converged with residents' environmental rights claims: in order to safeguard local ecological interests, it not only represented residents in appeals to higher-level authorities but also engaged in negotiations with neighboring governments. However, due to administrative fragmentation and the constraints of territorially bounded governance, these claims failed to generate effective regulatory intervention. Consequently, rights-based claims gradually shifted from institutional expression toward more confrontational forms of collective action. Symbolic protests—such as dumping dead fish in front of the Shengze Town government—sought to

render environmental degradation both visible and politically salient. When such actions proved insufficient, affected communities further escalated their response by blocking the Qingxi River, directly intervening in the material flows of pollution. The escalation of this dispute functioned as a mechanism for pressure transmission within China's hierarchical governance system. As the conflict intensified and attracted broader public and media attention, it generated upward accountability pressures, prompting intervention by higher-level authorities. The subsequent involvement of provincial and central government agencies transformed a localized dispute into a cross-regional governance issue, necessitating a coordinated institutional response.

In 2001, Jiangsu Province and Zhejiang Province held a coordination meeting on prevention and resolution of inter-provincial water pollution disputes. The Ministry of Water Resources, Ministry of Ecology and Environment, and both provincial governments reached a mediation agreement. As part of the resolution, Shengze Town committed to improving its outflow water quality, ensuring that by the end of 2002, the permanganate index would meet Grade V standards, and by the end of 2003, it would reach Grade IV standards. On January 29, 2002, the National Wangjiangjing Water Quality Automatic Monitoring Station was officially launched. As part of the national monitoring system, it was recognized for providing accurate and up-to-date water quality data. This marked a turning point for the Qingxi River, as it was now under real-time supervision by the central government, ensuring strict monitoring of industrial pollution control and preventing excessive emissions. On February 7, 2002, the governments of Suzhou and Jiaxing held their first joint meeting on cross-boundary water pollution, marking the beginning of coordinated governance efforts. The two cities established a joint working group responsible for emergency response and rapid action. To enhance monitoring, Wujiang District and Xiuzhou District each set up water quality monitoring stations, where environmental protection departments regularly conducted sampling and testing. The monitoring results indicated that the water quality remained at Grade V, reflecting severe pollution.

The conflict triggered by industrial pollution forced the Jiangsu authorities to take steps to abate the Qingxi River's pollution. The shared governance goals create a bridge between the two towns. Under a routine governance mode—referring here to an environmental management approach operating in the absence of acute crises and relying on periodic monitoring, information sharing, and jurisdiction-based accountability—the two local governments conducted regular inspections, exchanged environmental data, and engaged in negotiations when pollution incidents were detected. These practices contributed positively to the reduction of transboundary water pollution. This phase was characterized by each town being primarily responsible for its own environmental management.

#### 4.3. Local development direction from growth machine to environmental constraint

After the reform and opening-up, the decentralization of local fiscal authority and the GDP-driven performance evaluation system incentivized local governments to attract capital through policy innovations, while enterprises leveraged local protection to circumvent environmental regulatory costs. This formed a closed-loop interest structure of "growth-political achievement-revenue," known as the "growth coalition." Shengze Town government strictly controlled land use rights, monopolized land supply, and utilized low tax rates and subsidized industrial land prices to attract investment. By creating a business-friendly institutional environment, the local government prioritized industrial development, which led to a boom in the textile industry. For instance, in 2000, there were 697 enterprises in Shengze with 11,344 shuttleless looms, an increase of 6868 compared with 1999.

Compared to Wangjiangjing Town and other nearby areas, textile factories have preferred Shengze Town as an industrial production base. It is because Shengze's textile industry chain is complete, and the cost of



pollution control is relatively low, creating favorable conditions for capital accumulation. In 1992, Shengze accounted for approximately one-sixth of China's annual exports of real silk, and its township and village industrial output ranked first nationwide, with eight villages with output values exceeding 100 million yuan. Under the goal of economic growth, a competitive market for industrial products has been formed between the local government and industrial and commercial entities in a solid coalition through land and taxes. This led to a spatial pattern characterized by intermingled industrial and residential zones. However, the rapid industrial expansion outpaced the construction of environmental infrastructure, particularly wastewater treatment facilities. Large quantities of untreated industrial wastewater were discharged directly into the northern waters of Zhejiang, exacerbating environmental degradation. As a direct consequence, the water quality of Qingxi River deteriorated to below Grade V, and residents downstream in Wangjiangjing Town lived in a heavily polluted environment. Their livelihoods were disrupted, and even some villages were afflicted with elevated cancer rates. The primary economic benefits of this development were captured by the growth coalition, while residents of both towns bore the burden of environmental pollution.

In pursuit of a common environmental goal, fishermen spearheaded an anti-growth coalition that combined spontaneous mobilization with organized collective action. The “Grassroots Midnight Action” disrupted the previously stable coalition between the local government and industrial enterprises. The scale of the transboundary pollution issue was elevated, symbolizing the intervention of the central government's public power at the highest level. This forced the two local governments to change their governance approach to cope with top-down pressure. Firstly, the head of the Environmental Protection Bureau of Wujiang District was removed from his post, which sent a strong warning signal to local officials and created a deterrent effect. Second, the immense environmental pressure from the daily discharge of 100,000 tons of sewage changed the interest structure between local government and industrial enterprises. In response, a strict pollution control strategy was implemented by dividing 25 enterprises into five management groups, each further split into two subgroups, with alternating production days. This measure effectively reduced total emissions to about 40% of the previous levels. Factories that failed to comply with the alternating production schedule, exceeded pollution limits, or violated emission restrictions faced penalties, including a seven-day shutdown. Moreover, wastewater treatment was centralized, with all industrial effluents processed collectively. The cost of wastewater treatment was based on both the volume and concentration of discharge, incentivizing enterprises to reduce pollution. This restructuring significantly altered Shengze's industrial landscape. Factories could only operate one day and had to shut down for the next two, drastically reducing production time. Additionally, mandatory pollution fees added financial pressure, making it difficult for small textile and dyeing enterprises to sustain operations, leading many to shut down or relocate. Meanwhile, larger firms were forced to upgrade their equipment and processes, phasing out outdated water-jet looms and adopting cleaner technologies. As a result, Shengze experienced an industrial upgrade, shifting toward a sustainable development. This major public incident heightened both governments' focus on the environment and strengthened residents' environmental awareness and willingness to cooperate. This laid the groundwork for future cross-regional environmental governance collaboration.

The “Grassroots Midnight Action” attracted broader political and public attention, transforming what had previously been fragmented and localized pollution disputes into a cross-administrative governance issue. This process of “scale jumping” triggered vertical bureaucratic accountability and prompted intervention by higher-level authorities. It was within this context that institutional innovations—such as the joint river chief system—emerged. These arrangements should not be understood as isolated policy initiatives, but rather as responses to escalating governance pressures generated by bottom-up contention.

## 5. Green development coalition based on the joint river chief system

### 5.1. Growing cross-boundary cooperation under ecological civilization strategy

The “Grassroots Midnight Action” exposed the escalating severity of transboundary water pollution and articulated residents' urgent demands for environmental governance. It compelled governments at multiple levels to confront the systemic failures of cross-boundary governance and the burgeoning ecological crisis, thereby catalyzing the search for durable, institutionalized policy instruments. Since 2012, the “Ecological Civilization” has been one of the core strategies within China's national governance system. It has elevated environmental protection from a social issue to a national strategy and political task. Wujiang District in Suzhou, Jiangsu Province, and Xiuzhou District in Jiaxing, Zhejiang Province, are geographically adjacent and interlaced with dense water networks. Taking Shengze Town as an example, it has 392 rivers and lakes, with a total waterway length of about 333.34 km. The border area between Shengze Town and Zhejiang's Xiuzhou District spans approximately 27 km. In most cases, Shengze Town is located upstream of Xiuzhou District, and the Qingxi River is the most important boundary river between the two areas. As a result, the water quality in Wujiang District directly affects Xiuzhou District. Since 2012, Xiuzhou District has proactively worked with Wujiang District to break down administrative barriers and overcome the challenges of cross-provincial water governance. On June 13, 2012, the policy document “Guiding Opinions on the Establishment of the Inter-Provincial Boundary Environmental Joint Law Enforcement Working Mechanism between Xiuzhou District and Wujiang District” was issued, initiating mutual inspections of industrial wastewater discharge within a 5-km zone on both sides of the Qingxi River. This agreement has enhanced communication and cooperation between the two regions.

The river chief system was first proposed by the municipal government of Wuxi as a response to blue algae blooms in Taihu Lake in 2007. This local initiative was then promoted by the Jiangsu provincial government to other areas of the Taihu Basin and eventually elevated to a national water governance strategy. It is one of the key institutional outcomes of the ecological civilization strategy. In 2016, the Central Committee of the Communist Party of China and the State Council issued the “Opinions on Comprehensively Implementing the River Chief System”, advancing the river chief system reform to tackle China's water problems. This policy document, with its high authority and effectiveness, provided a powerful impetus for the system's rapid nationwide implementation. The river chief system mandates that main leaders within the Communist Party or government of China at the provincial, city, county, or township level serve as river chiefs. With the adoption of the river chief system, the performance assessment system now includes evaluations of local water management. The enforcement and regulatory work include six key tasks, such as strengthening the protection of river and lake shorelines and preventing water pollution. While this policy document clearly defines the overall goals and tasks of the river chief system, some tasks, such as quantitative and phased targets, are intentionally left flexible in the central government's documents. As a result, the system's effectiveness depends more on the local governments' implementation and innovation. As a governance approach based on the principle of territorial management, the river chief system has demonstrated considerable effectiveness within a single administrative jurisdiction. However, its limitations become evident when applied to transboundary rivers. On the one hand, officials at the same administrative level lack hierarchical subordination, making it difficult to establish an effective vertical chain of command. On the other hand, in the absence of coordination and oversight from higher-level authorities, the governance of transboundary rivers lacks stable institutional guarantees.

Wujiang District and Xiuzhou District are situated downstream of the

Taihu Lake—China's third-largest freshwater lake—and at one of its primary outlets. The water quality in this region is therefore vital to the health and water security of tens of millions of people. Following the severe blue algae blooms in 2007, the remediation of Taihu Lake was elevated to a national strategic priority. As the directly responsible jurisdictions, Wujiang and Xiuzhou have since been under immense pressure from central and provincial governments to meet stringent performance assessments. Historically, as two of China's pioneering industrialized regions and renowned hubs for the textile industry, these districts were major contributors to the lake's industrial pollution. They have experienced prominent transboundary environmental pollution incidents, thus receiving a high degree of political attention.

The hydrological connectivity between the Wujiang and Xiuzhou presents significant challenges for waterway maintenance. These challenges are compounded by the unique characteristics of the navigation channels and river courses, their extensive transit lengths, and particularly the large-scale seasonal proliferation of water hyacinth, which severely impacts the interconnecting rivers and results in a substantial maintenance workload. In 2017, Xiuzhou and Wujiang signed an Agreement on Joint Prevention and Control of Water Environment in Border Areas, establishing a coordinated mechanism that included organizational structures, information sharing, and regular inspections. The two regions established a monthly notification system for water quality monitoring data in their border areas. Based on the principle of territorial jurisdiction, each locality's responsibility for water environment sanitation was clearly demarcated. Furthermore, joint law enforcement operations were to be conducted at least once per quarter. A notable instance of collaboration occurred in September 2017, when an influx of upstream water hyacinth overwhelmed the cleaning capacity of Shengze Town's existing vessels and personnel. In response, Wangjiangjing Town provided three automated salvage boats to assist Shengze in managing the crisis. Subsequently, in 2018, Wangjiangjing provided technical guidance and design drawings to help Shengze construct its own automated water hyacinth harvesting station. This development significantly alleviated the pressure of water hyacinth control downstream. These cooperative efforts signify the progressive establishment of trust and collaboration in water governance between Wujiang and Xiuzhou.

## 5.2. Joint river chief system with multi-level and multi-subject participation

The transboundary cooperation agreements signed between Wujiang District and Xiuzhou District primarily focused on single issues and were executed by functional departments such as the water conservancy and environmental protection. A drawback of this form is the high cost of coordination when higher-level decisions or inter-departmental collaboration are required. Since it involves two different provinces, every step from project initiation to approval had to be reported up through multiple administrative levels, resulting in low efficiency. In response, Wangjiangjing Town proposed the establishment of a long-term cooperation mechanism and submitted the idea to Xiuzhou's Water Management Office, which gained the support of the Xiuzhou District Government. After consultation with Wujiang District, the two governments decided to pilot a "joint river chief system."

The joint river chief system establishes a hierarchy of district-level, town-level, and village-level river chiefs, with leaders within the Communist Party or government serving in these roles. District-level river chiefs are responsible for major rivers and important bodies of water within their jurisdiction, with a particular focus on rivers that cross district boundaries. They are tasked with coordination of river and lake management across districts, developing policy plans, resolving water environment issues that span towns and subdistricts, and coordinating resources like funding, engineering projects, and law enforcement. Town-level river chiefs primarily oversee important tributaries and medium-to-small rivers and lakes within their administrative areas,

especially sections that cross town or village borders. They are responsible for translating district-level directives into detailed town-level governance plans and for guiding and inspecting the work of village-level river chiefs. They also need to conduct joint patrols and enforcement with neighboring towns to resolve disputes that cross village lines. Village-level river chiefs, typically held by village party secretaries, are responsible for rivers, small ditches, and ponds within their villages. Their duties include regular river patrols, mobilizing residents to participate in river protection, reporting issues, and cooperating with higher-level departments. Compared to the river chief system, the joint river chief system emphasizes the management of water resources at the watershed level and the integration of governance functions across regions.

Wujiang District and Xiuzhou District have divided their joint river chiefs' works into five key aspects (Table 4). These include joint river patrols, river channel remediation, water quality monitoring, water environment cleaning, and law enforcement. On June 12, 2018, during a joint patrol, the town-level river chiefs of Wujiang and Xiuzhou discovered severe silt accumulation in the Qingxi River. After consultations, the two sides signed a cooperative agreement for joint governance of the Qingxi River and decided to invest 100 million yuan to dredge and desilt the watercourse. Within six months, the silt that had accumulated over 40 years was completely cleared. Wujiang was responsible for 6.95 km of the river, with a dredging volume of 598,000 cubic meters, while Xiuzhou handled 3.515 km, dredging 408,500 cubic meters. This project is regarded as a milestone in the implementation of the joint river chief system. A lifelong local resident recalled: *"It was not always like this. The river used to be covered in layers of water hyacinths, the water was murky and foul-smelling, and no one dared to open their windows."* (Interviewee WV3) *"Residents from both sides used to blame each other."* (Interviewee WV4)

On November 27, 2018, the two districts held a Joint River Chief Appointment Ceremony, officially recognizing this cross-jurisdictional governance model. A total of 58 river chiefs were appointed, covering 18 boundary rivers and lakes. An official from Wangjiangjing Town explained: *"We patrol the river once a week. We have to check water quality and identify any anomalies (...) There are three levels of river chiefs—village, township, and district—each with designated personnel in charge. Villages are also required to conduct frequent patrols. The information must be uploaded to an application, and if anyone fails to complete their patrol, a public notice will be issued."* (Interviewee WG3) River chiefs are required to patrol regularly, document river conditions with photographs, and upload them to a designated application. Failure to conduct inspections within the prescribed time frame results in public reprimand. Past

**Table 4**  
Key mechanisms of the joint river chief system.

| Mechanism             | Content                                                                                                                                                                                                                                  |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Joint River Patrols   | River chiefs at various levels conduct regular joint patrols. Issues are discussed and resolved on-site in real time.                                                                                                                    |
| Joint Monitoring      | Standardized monitoring points, indicators, schedules, and evaluation methods are established. Environmental monitoring data and major water safety information are shared across regions.                                               |
| Joint Law Enforcement | A specialized enforcement patrol team is formed with professionals from neighboring areas. Joint law enforcement patrols are conducted regularly. Inspection results, notifications, and solutions are jointly reviewed and implemented. |
| Joint Cleaning        | A unified river and lake maintenance system is developed. Standardized cleaning equipment and protocols ensure consistent quality. Regular joint inspections, consultations, and debris removal efforts are carried out.                 |
| Joint Governance      | Collaborative planning for river restoration, including scale, investment, content, and timelines. Synchronization of measures across the upstream, downstream, and both banks of the river.                                             |

environmental pollution had caused unimaginable damage, sparking widespread public concern over drinking water safety and health risks. Consequently, residents developed strong environmental and rights-protection awareness. *"The environmental awareness of citizens is remarkably strong. If the color of a river changes, it usually indicates that a factory is responsible—either through illegal wastewater discharge or a pipeline malfunction. Once residents notice this, they will report it by phone."* (Interviewee WG1) *"I would say that environmental awareness among residents in Zhejiang is among the strongest in the country. In other provinces, residents might not approach the Environmental Protection Bureau even if they smell odors or see polluted waterways; however, residents in Zhejiang know how to use institutional channels to file complaints and safeguard their interests."* (Interviewee WG2). In Wangjiangjing, there were cases in which residents reported enterprises for illegal wastewater discharge, leading to fines and even criminal convictions of company executives. Through routine supervision and problem reporting, grassroots communities provide an important supplement to formal government-led governance.

### 5.3. Grassroots efforts in cross-provincial exploratory cooperation

The joint river chief system has reorganized and institutionalized grassroots participation in the governance of transboundary rivers. Prior to the establishment of formal governance arrangements, local residents primarily engaged in environmental governance through petitions, protests, or collective action. While these actions exerted a degree of pressure, they were often intermittent and difficult to sustain. With the implementation of the joint river chief system, grassroots participation has gradually been incorporated into a more structured governance framework. Village cadres, elected by villagers and occasionally serving as village-level river chiefs, perform a pivotal intermediary role, linking higher-level administrative authorities with local communities. Their responsibilities extend beyond policy implementation to include coordination, communication, and social mobilization. At the same time, local residents have shifted from being primarily claim-makers to active participants in routine governance practices, such as environmental monitoring, reporting, and collective maintenance activities. In this sense, grassroots social pressure has not been eliminated; rather, it has been absorbed and reconfigured within a state-led governance structure.

Xiyan Village in Wangjiangjing Town and Xingqiao Village in Shengze Town are adjacent, located on the border between Zhejiang and Jiangsu provinces. The Xiaotianwei River, which serves as both the boundary between the two villages and an inland section of the inter-provincial Qingxi River, marks their shared border. The two villages lie in a peri-urban zone where agricultural, industrial, and residential spaces are interwoven, characterized by a high density of migrant populations. For instance, Xingqiao contains more than 18,000 square meters of factory buildings and has a population exceeding 2600. In the past, local villagers relied on a single bridge for passage. To facilitate crossing, they piled up an earthen dam, which obstructed water flow. Meanwhile, villagers along the banks constructed sheds and raised livestock and poultry, causing the water quality to steadily decline until it reached Grade V—the lowest grade in China's surface water standards—turning the river into a "black and odorous" waterway.

The party secretaries of Xiyan and Xingqiao, serving as village-level river chiefs under the joint system, are required to conduct regular patrols of the Xiaotianwei River. *"We typically patrol the river four to five times a month."* (Interviewee WC1) *"By identifying and resolving issues together, we avoid multi-layered communication, which significantly improves our work efficiency."* (Interviewee SC1) The regular joint patrols have fostered a closer relationship between the two villages. In the second half of 2022, while on a joint patrol, the two secretaries conceived the idea of remediating the Xiaotianwei River. However, the human and material resources required exceeded the combined capacity of the two villages. As a result, they reported the issue to their respective township governments, effectively shifting the governance scale

upward.

Remediating Grade V rivers is a major policy objective of Zhejiang province's water management initiative. For the township government, this is a significant political and public welfare task tied directly to official performance evaluations. As one local official recalled: *"The river's water quality was very poor for several consecutive months (...) Located in a peri-urban area, the population is complex and involves a large number of people. Other villages are fine, but this village's problems are the most prominent."* (Interviewee WG1) Compared to past collaborations on the Qingxi River—which mainly involved cleaning and dredging—the scale of this new project was far more challenging. It required village-level governance that involved local residents, and it even necessitated the removal of their vegetable plots and sheds. There was no precedent for this type of cross-township cooperation. Wangjiangjing Town officials, seeking a solution, decided to reach out to Shengze Town. An official from Wangjiangjing stated, *"At the time, we were just exploring and didn't have high hopes (...) We went to Shengze, discussed it with their River Chief Office, and they said they were willing to do it, and we said we were willing to do it. We decided to invest some money."* (Interviewee WG3) Shengze Town agreed to Wangjiangjing's proposal. For Shengze, despite its strong economic power—with a GDP comparable to that of a county-level city—it faced imbalances between industrial growth and service provision. Such developmental disorder constrained its long-term prospects, compelling officials to improve ecological and environmental quality, including in rural areas, alongside economic expansion. As one Shengze official noted: *"Wangjiangjing officials came to us (to seek cooperation) because of poor environmental conditions. Their environmental protection bureau took the lead, and on our side the River Chief Office was in charge. (...) At the end of 2022, during the annual River Chief meeting, I raised the idea with the Water Affairs Bureau: we should select a boundary river with Zhejiang for joint remediation, to create a project at the village level, so that more residents in surrounding areas could benefit."* (Interviewee SG3) Thanks to the strong industrial bases of both Wangjiangjing and Shengze—the Xiaotianwei remediation project received robust financial backing. This enabled the initiative to embody the principle of "industry supporting agriculture," channeling industrial prosperity into rural environmental improvement.

Although the joint river chief system had already been established in Shengze Town and Wangjiangjing Town at that time, the specific implementation of the Xiaotianwei project remained constrained by existing administrative divisions and territorial management. The two localities were separately responsible for the full set of procedures within their respective jurisdictions, including project proposal submission, tendering and implementation. Wangjiangjing Town reported the project as a simplified engineering project, with a relatively straightforward approval process and funding drawn from the township-level budget; by contrast, Shengze Town incorporated it into its "Sub-standard Grade V River Remediation Project," which involved a more complex approval procedure. This contrast reflects the administrative autonomy and flexibility of local governments, while also revealing the reality that fiscal responsibilities and administrative powers remain highly localized in cross-boundary governance. Similar fragmentation was also evident in project tendering and technical implementation. The two governments organized separate bidding processes and oversaw construction within their respective territories; technical standards and implementation procedures were also determined independently. While they commissioned professional planning and design teams and communicated on architectural styles to maintain visual consistency, discrepancies persisted in design details and construction pacing. Wangjiangjing Town completed its tendering in the first half of 2023, reaching readiness for construction, whereas Shengze Town remained at the approval stage. To avoid further disruption to the lives of riverside villagers caused by unsynchronized construction, Wangjiangjing Town voluntarily delayed its construction schedule to align with Shengze Town's timeline. This seemingly cooperative outcome did not stem from unified scheduling under institutional integration; rather, it was

achieved through informal consultation and the unilateral concession of time costs. The subsequent operation and maintenance also adhered to territorial responsibility, with no cross-regional joint maintenance mechanism established. Overall, cross-boundary coordination did not eliminate institutional fragmentation; instead, it operated primarily through negotiation and compromise.

On September 19, 2023, the groundbreaking ceremony for the Xiaotianwei Project was officially held. The major construction tasks included the following (Table 5).

These efforts significantly improved the river's landscape and water quality (Fig. 2). The previously silted-up river channel has been restored to unobstructed flow with marked improvement in water quality. Five pedestrian boardwalks have been constructed to facilitate residents' cross-river mobility. Villagers now regularly utilize the riverside fitness trails for walks during leisure hours. *"The whole area has become park-like with its improved environment."* noted Interviewee SV2. *"With these new bridges, getting to the opposite bank has never been easier!"* (Interviewee WV5) However, these improvements were not achieved without cost. Two enterprises were shut down during the environmental remediation process, resulting in a reduction in collective village income and the loss of employment for some villagers. For those affected, the public environmental benefits derived from ecological restoration did not fully offset the losses sustained in terms of income and livelihoods. In this sense, environmental improvement cannot be understood as a unidirectional process of public interest realization. Rather, the restoration process was accompanied by a reconfiguration of interests, risks, and responsibilities among various stakeholders.

The stability of the green development coalition does not stem solely from intergovernmental coordination; rather, it is grounded in the strategic alignment of multiple stakeholders. The grassroots level is where the demand for cross-boundary governance and the willingness for regional cooperation are most keenly felt. In the Xiaotianwei River project, the leadership of village cadres played a crucial role, including their strong sense of public service, decision-making capacity, and organizational mobilization skills. As one official noted, *"The village committee was the most involved in this work because we are mainly responsible for construction, and we need their assistance to coordinate with the villagers and persuade them."* (Interviewee SG2) Village cadres, acting as intermediaries between the state and society, mobilize villagers—who might otherwise remain passive observers—by leveraging informal social networks, such as kinship ties and local customs, to engage them in land coordination and long-term river maintenance. This bottom-up social embeddedness ensures that governance reflects community "rights," rather than merely administrative "power." Despite being in different provinces, the villagers of the two communities share geographical proximity, kinship ties, shared language, and similar

customs, fostering a strong sense of social and cultural identity. The very process of jointly managing Xiaotianwei further reinforced this shared identity, generating new forms of social linkage. Currently, this project has become a model for integrated regional development, garnering significant media attention from influential outlets. Village cadres sought to leverage media exposure to raise the area's visibility, gain recognition from higher-level authorities, and strengthen their competitiveness in future evaluations for honors, pilot programs, or subsequent allocations of funding. For local actors, such symbolic recognition or potential material rewards constitute a vital source of incentives for participating in transboundary cooperation. At the same time, media coverage and public attention amplify local environmental issues beyond administrative boundaries, triggering broader public scrutiny and reputational risks. Importantly, these incentive mechanisms are not limited to governmental actors. Enterprises, under the combined pressures of regulatory enforcement, public oversight, and reputational concerns intensified by media exposure, are compelled to invest in pollution control, upgrade production processes, and comply with environmental standards. Meanwhile, residents and grassroots organizations are not merely passive beneficiaries but active participants, contributing through reporting mechanisms, local coordination, and ongoing monitoring, thereby embedding social accountability into the governance process. This dynamic elucidates the operation of a multi-stakeholder alliance comprising local governments, enterprises, grassroots organizations, and broader societal actors. The sustainability of this coalition depends on the alignment of incentive structures across participants. When policy enforcement weakens, compensatory arrangements are absent, or symbolic rewards fail to materialize, the incentives for sustained engagement among local governments, residents, and social organizations may decline, increasing the risks of strategic compliance or governance fatigue. By contrast, when ecological outcomes are translated into tangible developmental or political returns—such as access to funding, enhanced reputation, or the establishment of inter-jurisdictional cooperation mechanisms—the coalition is more likely to achieve enduring stability beyond cyclical policy campaigns.

#### 5.4. Governance transformation from growth coalition to green development coalition

In the past, China implemented a development strategy centered on "economic construction." Through fiscal decentralization and a cadre evaluation system, a GDP-oriented local competition regime was established. The Shengze Town Government provided land, loans, and policy support, while enterprises contributed taxes and employment, relying on local protectionism to evade environmental regulations and compliance costs. Together, they formed a growth coalition. Industrialization brought significant economic benefits to Shengze Town, but the accompanying industrial wastewater was transferred downstream, causing pollution in Wangjiangting Town. It intensified competition—and even conflict—between the two local governments. However, when pollution accumulated to a critical threshold and catalyzed collective action among villagers, the growth machine faced a process of social regulatory correction. This bottom-up social pressure that attracted vertical intervention from central authorities, disrupted the equilibrium of the local growth coalition, and compelled local governments to incorporate environmental restoration into their policy agendas.

As the central government incorporated environmental governance into the national strategic framework, it strengthened the legal and regulatory system for environmental protection and embedded environmental indicators into the cadre management and performance evaluation system. At the same time, through the deployment of central environmental inspections as a supervisory instrument and the designation of ecological protection zones to restrict economic development, the state has transformed environmental governance from a social issue

**Table 5**  
Construction work of Xiaotianwei River.

| Position           | Construction work                                                                                                                                                                                                          | Investment amount |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Shengze Town       | 10,100 cubic meters of river dredging<br>4 new wooden bridges<br>803.3 meters of new ecological revetment<br>564 square meters of pedestrian walkways<br>3213.2 square meters of new green spaces                          | 3 million yuan    |
| Wangjiangting Town | 9100 square meters of river improvement<br>730 square meters of small pond rehabilitation<br>1040 meters of new ecological revetment<br>656 square meters of pedestrian walkways<br>1300 square meters of new green spaces | 2 million yuan    |





Fig. 2. Changes of Xiaotianwei river.

Sources: Pictures provided by Xingqiao Village Committee. Images a1–a3 and b1–b3 respectively show the same locations before, during, and after the Xiaotianwei River project.

into a clearly defined political task, significantly elevating its salience on local policy agendas. The widespread implementation of the river chief system is a direct result of these top-down efforts. Local governments are now compelled to allocate funds and administrative resources to areas such as ecological restoration and green infrastructure development, often outside of their traditional economic development agendas.

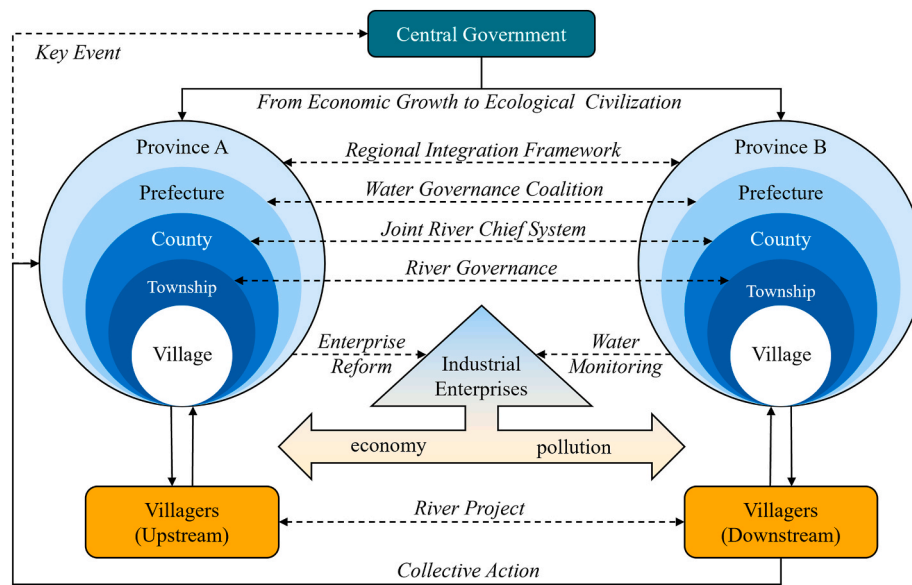
Simultaneously, the central government has guided and regulated interregional cooperation among local governments through various methods, including development plans, cross-provincial coordination mechanisms, and legal and policy safeguards. The elevation of the Yangtze River Delta integration to a national strategy not only reinforced the legitimacy of cross-boundary collaboration at the institutional level but also reshaped interlocal relations in practice, making cross-boundary cooperation an intersection of local interests and national will. Within this context, the emergence of the joint river chief system became institutionally and politically feasible. However, this does not mean that the system is purely a top-down policy product. Its formation is more a response to the complex and multifaceted environmental issues that necessitate cooperative governance. It can thus be seen as a practice of institutional innovation at the local level. Local governments, exemplified by Wangjiangjing, have actively initiated cross-boundary cooperation proposals to higher-level authorities in response to real-world challenges. This led to higher-level governments taking the lead in institutionalizing cross-boundary governance through bilateral negotiations. This process highlights the agency and creativity of local actors in both cross-boundary governance and institutional innovation.

The joint river chief system assigns specific river chiefs to cross-boundary rivers, binding governance responsibilities to multiple levels of government across different jurisdictions. The hierarchical relationship of “coordination–implementation/reporting” runs throughout its operation: higher-level governments coordinate cross-boundary efforts and allocate resources, while grassroots governments and local communities are responsible for implementing the governance plans and reporting on local challenges. The institutionalized mechanism of horizontal coordination makes cross-boundary cooperation more stable, rather than dependent on ad hoc negotiations. This, in turn, enables the implementation of large-scale governance projects and legitimizes more flexible cooperation at the grassroots level. Regular activities such as river patrols and enforcement strengthen interaction between local communities across boundaries, thereby building a cross-boundary cooperation network centered on river governance.

Past environmental pollution incidents and the ensuing collective actions have significantly heightened residents’ environmental awareness. Whenever abnormalities in river water quality or instances of illegal industrial discharge are detected, residents frequently take the

initiative to report through multiple channels, such as village committees, township environmental departments, or public hotlines. Relevant authorities are then obliged to investigate and provide a response within a prescribed timeframe. This social supervision mechanism has, to some extent, imposed constraints on enterprises and exerted pressure on local governments. It aligns with the evolution of social movements from “confrontational” to “institutionalized.” It is precisely this sustained public concern and expectation for ecological improvement that has laid the social foundation for the collaborative governance of the Xiaotianwei River. Against this backdrop, enterprises have transcended the passive acceptance of environmental penalties. Instead, they proactively respond to governance mandates by investing in the green transformation of production processes, standardizing emissions, and cooperating with information disclosure. Certain key enterprises have even entered into collaborative development agreements with local villages, participating in rural planning and management through targeted donations, financial aid, and the provision of technical and managerial expertise. Village cadres—particularly the village party secretaries—often serve concurrently as “joint river chiefs.” They are responsible for communicating environmental issues and grassroots needs to higher-level governments to secure policy and resource support. At the same time, they must explain policies to villagers and mobilize them for local action. In this process, village cadres act as key nodes for policy transmission, interest coordination, and governance implementation. Furthermore, grassroots organizations such as village committees transform fragmented individual demands into institutionally absorbable governance issues through daily consultation and interest coordination, thereby functioning as a nexus between the state, enterprises, and residents. In this process, media coverage and public opinion provide essential support for grassroots organizations to garner external attention and resources, further strengthening multi-stakeholder interaction. In certain contexts, local environmental volunteers or informal social organizations enhance the social embeddedness of the governance process through patrols, monitoring, and negotiation. The Xiaotianwei project further illustrates the interaction between urban and rural dynamics. This is seen in how urban environmental targets and policy pressures are transmitted to rural areas, and how advanced ecological restoration technologies and cross-boundary governance models are applied and practiced within a rural ecological spaces.

Driven by the shared goal of environmental protection, multiple stakeholders—from local governments to village cadres and villagers—have gradually constructed a “green development coalition” that transcends administrative hierarchies and territories (Fig. 3). The formation of this coalition marks a shift in the motivation of local governance in China, from the singular pursuit of economic growth toward a more integrated goal encompassing ecological and social benefits. This



**Fig. 3.** Cross-boundary governance mechanism based on green development coalition.

Note: The blue circles represent local governments, with the four levels from largest to smallest corresponding to the provincial, prefectural, county, and township governments. Administrative units are distinguished using alphabetical labels. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

is not a simple case of “coalition replacement,” but rather a transformation in governance paradigms—from a “growth coalition,” characterized by interregional competition, economic expansion, and pollution-intensive development, to a “green development coalition,” oriented toward multi-scalar coordination, the revalorization of rural spaces, and green transformation. This shift not only constitutes a prominent feature of China’s urbanization and industrialization process but also reflects the country’s gradual evolution from a development path prioritizing efficiency toward one emphasizing equity and sustainability.

Fig. 3 presents an analytical framework for cross-boundary environmental governance that encompasses multiple scales, actors, and elements. It highlights how the central and local governments, enterprises, residents, and social organizations are embedded and coordinated across different administrative levels and spatial scales to promote the joint advancement of economic development and ecological restoration. The core structure of this mechanism is transferable insofar as transboundary environmental governance commonly confronts a mismatch between the operational scale of ecological systems and the scale of administrative management. This scale mismatch constitutes a shared governance challenge across regions. Against this backdrop, multi-level governance and cross-jurisdictional collaboration emerge as fundamental institutional responses to transboundary environmental problems, with governance practices typically exhibiting nested organizational structures. The coupling of top-down institutional constraints and bottom-up collective action represents a generally applicable mechanism for addressing tensions in transboundary governance. The former provides institutional guidance through strategic objectives and policy frameworks, while the latter facilitates implementation through local collective action and community participation. At the same time, the intensity of resource allocation, modes of grassroots mobilization, and the relative bargaining power among different actors remain highly contingent on specific contextual conditions. The Yangtze River Delta, for example, benefits from relatively abundant fiscal capacity and governance resources, which strengthen the implementation of cross-regional collaboration. However, such variation reflects differences in governance capacity and resource conditions rather than fundamental divergences in governance logic. The broader significance of the green development coalition does not lie in replicating specific policy

instruments or organizational models, but in its value as an analytical lens that reveals how diverse actors coordinate, negotiate, and cooperate around ecological objectives, and how dynamic balances are formed between ecological restoration and local development. This framework provides a comparative analytical basis for understanding cross-boundary environmental governance across different institutional contexts, while also clarifying the conditions under which it is applicable.

## 6. Conclusion and discussion

Compared with the growth machine and the green growth machine—both rooted in neoliberal market mechanisms and emphasizing land value appreciation—the concept of the green development coalition highlights ecological restoration without undermining local economic development, particularly in ecologically fragile areas such as urban fringes and regional borders. In the past, both Shengze Town and Wangjiangjing Town were dominated by the textile and dyeing industry. The connection of production factors like land, taxation, and labor formed a “growth coalition” between local governments and industrial enterprises, which was primarily focused on economic growth. However, the industrial wastewater generated by these companies caused pollution that damaged downstream Wangjiangjing Town, leading to continuous disputes between the two regions. An “anti-growth coalition,” represented by local fishermen, organized collective actions in protest. In response, local governments were compelled to control pollution to address pressure from the central government. As the central government integrated environmental governance into the national governance framework, its priority on the local agendas was elevated. The demand for cooperative governance, spurred by cross-boundary environmental issues, drove the two governments to establish a joint river chief system under the institutional frameworks of Ecological Civilization, Integrated Development of Yangtze River Delta, and the River Chief System. This arrangement built a cross-boundary cooperation network centered on river governance and facilitated grassroots initiatives such as the Xiaotianwei project. Local governments thus operated under dual pressures: the coercive force of national strategies and the spontaneous demands of residents. Local communities’ supervision and reporting filled the gaps in top-down formal governance. Driven by the shared goal of environmental protection, multiple

actors—from local governments to village cadres and villagers—jointly constructed a green development coalition that transcends administrative hierarchies and territories.

The case examined in this study is situated in China's Yangtze River Delta. As one of the earliest industrialized areas, its advanced industrial foundation and strong financial capacity provide crucial support for bottom-up institutional innovation, particularly in cross-boundary governance projects. This presents a certain challenge for less economically developed regions, where such resources may be more limited. The effective functioning of the green development coalition relies on multi-scale collaborative governance. This places higher demands on the governance capabilities of grassroots governments, especially in the context of cross-regional cooperation. In Western countries or other developing nations, social organizations often have a high degree of autonomy and social efficacy, enabling them to, to some extent, substitute for or supplement government functions. This offers an alternative potential path for cross-boundary environmental governance initiatives. Polycentric governance theory emphasizes collective action achieved through negotiation and self-coordination among multiple relatively autonomous decision-making centers, with a core concern for the decentralization of governing authority and the capacity for local self-organization (Ostrom, 2010a, 2010b). By contrast, this case exhibits a governance arrangement in which local governments play a leading role while enterprises, villages, and residents are incorporated into a shared governance process through institutional embedding. At the same time, environmental stewardship theory highlights the active responsibility of individuals and groups to care for and sustain ecosystems over the long term, often emerging from locally grounded practices rather than formal regulatory mandates (Chapin et al., 2010; Bennett et al., 2018). In this respect, the green development coalition represents a state-led attempt to institutionalize and rescale cross-administrative environmental management. While such an approach significantly reduces uncertainty in cross-boundary governance through centralized institutional integration, it also depends heavily on local governments' fiscal capacity and administrative mobilization resources, constraining its replicability in less-developed contexts. The decentralization of authority and local embeddedness emphasized in polycentric governance can enhance institutional adaptability and long-term resilience, while environmental stewardship perspectives suggest that governance arrangements can avoid excessive reliance on external administrative mobilization only when they cultivate a sense of responsibility and sustained engagement among local actors.

Environmental governance in China is neither solely driven by the capital-accumulation logic of the “growth machine,” nor does it fully conform to the state-led doctrine of “ecological civilization.” Rather, the relative weight of these two forces has shifted across different historical stages (Zhang and Wu, 2024). The tension between decentralization and centralization runs throughout the relationship between the central and local governments (Zhu et al., 2025). In the past, local governments often under-enforced regulations in order to shield enterprises. However, with the central government's adjustments to fiscal and incentive mechanisms and the intervention of environmental inspection teams, local governments have been compelled to accelerate their green transformation. This includes innovating pollution treatment technologies, adopting energy-saving equipment, and fostering the rise of green businesses to progressively mitigate the environmental impact of economic growth. Green finance, low-carbon technologies, and related industries are thus becoming crucial pathways for promoting sustainable development (Flammer, 2021; Wang et al., 2018). Citizen complaints and supervision play a vital role in promoting the effective implementation and dynamic optimization of environmental regulations. Establishing a long-term mechanism for grassroots participation requires institutionalizing “cooperative participation” so that citizens can continuously gain a sense of efficacy (seeing their reported issues resolved) and a sense of benefit (gaining tangible benefits from environmental improvements). At the same time, administrative border

areas carry the potential to serve as connective hubs between jurisdictions. Yet differences in dominant industries, economic capacity, and hierarchical power structures exacerbate the difficulties of cross-boundary cooperation. Cross-jurisdictional public affairs—such as environmental protection, transportation connectivity, and resource sharing—involve overlapping economic interests and social responsibilities. These issues require support and coordination from higher-level governments, as well as collaboration among functional departments at all levels.

## CRedit authorship contribution statement

**Jie Yu:** Conceptualization, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing. **Chao Ye:** Conceptualization, Investigation, Methodology, Project administration, Writing – review & editing. **Liang Zhuang:** Conceptualization, Investigation, Methodology, Visualization, Writing – review & editing.

## Declaration of interest Statement

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

## Acknowledgments

This study was supported by grants from the National Natural Science Foundation of China (No. 42471281 & 42201249) and Fundamental Research Funds for the Central Universities (2026ECNU-WLJC002). The insightful and constructive comments of the editors and reviewers are appreciated.

## Data availability

Data will be made available on request.

## References

- Anguelovski, I., Connolly, J.J.T., Cole, H., Garcia-Lamarca, M., Triguero-Mas, M., Baró, F., Martín, N., Conesa, D., Shokry, G., Pérez Del Pulgar, C., Ramos, L.A., Matheney, A., Gallez, E., Oscilowicz, E., Máñez, J.L., Sarzo, B., Beltran, M.A., Minaya, J.M., 2022. Green gentrification in European and North American cities. *Nat. Commun.* 13 (1), 3816. <https://doi.org/10.1038/s41467-022-31572-1>.
- Anguelovski, I., Connolly, J.J.T., Masip, L., Pearsall, H., 2018. Assessing green gentrification in historically disenfranchised neighborhoods: a longitudinal and spatial analysis of Barcelona. *Urban Geogr.* 39 (3), 458–491. <https://doi.org/10.1080/02723638.2017.1349987>.
- Ansell, C., Gash, A., 2008. Collaborative governance in theory and practice. *J. Publ. Adm. Res. Theory.* 18 (4), 543–571. <https://doi.org/10.1093/jopart/mum032>.
- Bennett, N.J., Whitty, T.S., Finkbeiner, E., Pittman, J., Bassett, H., Gelcich, S., Allison, E. H., 2018. Environmental stewardship: a conceptual review and analytical framework. *Environ. Manage.* 61 (4), 597–614. <https://doi.org/10.1007/s00267-017-0993-2>.
- Berronet, L., Villegas-Palacio, C., Botero, V., 2020. Vulnerability of rural communities to change in an ecosystem service provision: surface water supply. A case study in the northern Andes, Colombia. *Land Use Pol.* 97, 104737. <https://doi.org/10.1016/j.landusepol.2020.104737>.
- Brenner, N., 2004. *New State Spaces: Urban Governance and the Rescaling of Statehood*. Oxford University Press.
- Bulkeley, H., 2005. Reconfiguring environmental governance: towards a politics of scales and networks. *Polit. Geogr.* 24 (8), 875–902. <https://doi.org/10.1016/j.polgeo.2005.07.002>.
- Bulkeley, H., Betsill, M., 2005. Rethinking sustainable cities: multilevel governance and the 'urban' politics of climate change. *Environ. Polit.* 14 (1), 42–63. <https://doi.org/10.1080/0964401042000310178>.
- Chapin III, F.S., Carpenter, S.R., Kofinas, G.P., Folke, C., Abel, N., Clark, W.C., Olsson, P., Stafford Smith, D.M., Walker, B., Young, O.R., Berkes, F., Biggs, R., Grove, J.M., Naylor, R.L., Pinkerton, E., Steffen, W., Swanson, F.J., 2010. Ecosystem stewardship: sustainability strategies for a rapidly changing planet. *Trends Ecol. Evol.* 25 (4), 241–249. <https://doi.org/10.1016/j.tree.2009.10.008>.
- Checker, M., 2011. Wiped out by the “greenwave”: environmental gentrification and the paradoxical politics of urban sustainability. *City Soc.* 23 (2), 210–229. <https://doi.org/10.1111/j.1548-744X.2011.01063.x>.
- Chen, G., Xu, J., Qi, Y., 2022. Environmental (de) centralization and local environmental governance: evidence from a natural experiment in China. *China Econ. Rev.* 72, 101755. <https://doi.org/10.1016/j.chieco.2022.101755>.



- Chien, S.S., Hong, D.L., 2018. River leaders in China: Party-state hierarchy and transboundary governance. *Polit. Geogr.* 62, 58–67. <https://doi.org/10.1016/j.polgeo.2017.10.001>.
- Comby, E., Le Lay, Y.F., Piégay, H., 2019. Power and changing riverscapes: the socioecological fix and newspaper discourse concerning the Rhône River (France) since 1945. *Ann. Am. Assoc. Geogr.* 109 (6), 1671–1690. <https://doi.org/10.1080/24694452.2019.1580134>.
- Dilworth, R., Stokes, R., 2013. Green growth machines, LEED ratings and value free development: the case of the Philadelphia property tax abatement. In: Urban, J. (Ed.), *Int. Res. Placemaking Urban Sustain.* 6 (1), 37–51. <https://doi.org/10.1080/17549175.2012.692570>.
- DuPuis, E.M., Greenberg, M., 2019. The right to the resilient city: progressive politics and the green growth machine in New York City. *J. Environ. Stud. Sci.* 9, 352–363. <https://doi.org/10.1007/s13412-019-0538-5>.
- Ekers, M., Prudham, S., 2017. The metabolism of socioecological fixes: capital switching, spatial fixes, and the production of nature. *Ann. Am. Assoc. Geogr.* 107 (6), 1370–1388. <https://doi.org/10.1080/24694452.2017.1309962>.
- Feng, Y., Wu, F., Zhang, F., 2024. Environmental statecraft and changing spatial politics: Erhai Lake protection in China. *Polit. Geogr.* 115, 103196. <https://doi.org/10.1016/j.polgeo.2024.103196>.
- Flammer, C., 2021. Corporate green bonds. *J. Financ. Econ.* 142 (2), 499–516. <https://doi.org/10.1016/j.jfineco.2021.01.010>.
- Green, F., Stern, N., 2017. China's changing economy: implications for its carbon dioxide emissions. *Clim. Policy* 17 (4), 423–442. <https://doi.org/10.1080/14693062.2016.1156515>.
- Gu, Y., Wu, Y., Liu, J., Xu, M., Zuo, T., 2020. Ecological civilization and government administrative system reform in China. *Resour. Conserv. Recycl.* 155, 104654. <https://doi.org/10.1016/j.resconrec.2019.104654>.
- Guttman, D., Young, O., Jing, Y., Bramble, B., Bu, M., Chen, C., Furst, K., Hu, T., Li, Y., Logan, K., Liu, L., Price, L., Spencer, M., Suh, S., Sun, X., Tan, B., Wang, H., Wang, X., Zhang, J., Zhang, X., Zeidan, R., 2018. Environmental governance in China: interactions between the state and “nonstate actors”. *J. Environ. Manage.* 220, 126–135. <https://doi.org/10.1016/j.jenvman.2018.04.104>.
- Han, X., Sheng, J., 2024. Governing the future through ‘ecological civilization’: anticipatory politics and China's Great Yangtze River Protection Programme. *J. Contemp. China* 33 (149), 774–789. <https://doi.org/10.1080/10670564.2023.2232747>.
- Hansen, M.H., Li, H., Svarverud, R., 2018. Ecological civilization: interpreting the Chinese past, projecting the global future. *Glob. Environ. Change, Hum. Pol. Dimens.* 53, 195–203. <https://doi.org/10.1016/j.gloenvcha.2018.09.014>.
- Harvey, D., 1981. The spatial fix—Hegel, von Thunen, and Marx. *Antipode* 13 (3), 1–12. <https://doi.org/10.1111/j.1467-8330.1981.tb00312.x>.
- Harvey, D., 1989. From managerialism to entrepreneurialism: the transformation in urban governance in late capitalism. *Geogr. Ann. Ser. B-Human Geogr.* 71 (1), 3–17. <https://doi.org/10.1080/04353684.1989.11879583>.
- Harvey, D., 2001. Globalization and the spatial fix. *Geographische revue* 2 (3), 23–31.
- He, G., Zhang, L., Mol, A.P.J., Wang, T., Lu, Y., 2014. Why small and medium chemical companies continue to pose severe environmental risks in rural China. *Environ. Pollut.* 185, 158–167. <https://doi.org/10.1016/j.envpol.2013.10.041>.
- Hooghe, L., Marks, G., 2003. Unraveling the Central State, but how? Types of multi-level governance. *Am. Polit. Sci. Rev.* 97 (2), 233–243. <https://doi.org/10.1017/S00030554030>.
- Hou, Y., Zhang, K., Zhu, Y., Liu, W., 2021. Spatial and temporal differentiation and influencing factors of environmental governance performance in the Yangtze River Delta. *China. Sci. Total Environ.* 801, 149699. <https://doi.org/10.1016/j.scitotenv.2021.149699>.
- Jiang, R., Lin, G.C.S., 2021. Placing China's land marketization: the state, market, and the changing geography of land use in Chinese cities. *Land Use Pol.* 103, 105293. <https://doi.org/10.1016/j.landusepol.2021.105293>.
- Jocoy, C.L., 2018. Green growth machines? Competing discourses of urban development in Playa Vista, California. *Urban Geogr.* 39 (3), 388–412. <https://doi.org/10.1080/02723638.2017.1333238>.
- Johansson, J., Holmquist, M., 2024. LEADER and rural development policy-what's the problem represented to be? *J. Rural Stud.* 108, 103287. <https://doi.org/10.1016/j.jrurstud.2024.103287>.
- Jonas, A.E., Wilson, D. (Eds.), 1999. *The Urban Growth Machine: Critical Perspectives, Two Decades Later*. Suny press.
- Kostka, G., 2016. Command without control: the case of China's environmental target system. *Regul. Gov.* 10 (1), 58–74. <https://doi.org/10.1111/rego.12082>.
- Kostka, G., Nahm, J., 2017. Central-local relations: recentralization and environmental governance in China. *China Q.* 231, 567–582. <https://doi.org/10.1017/S0305741017001011>.
- Lang, S., Rothenberg, J., 2017. Neoliberal urbanism, public space, and the greening of the growth machine: new York City's High Line park. *Environ. Plan. A.* 49 (8), 1743–1761. <https://doi.org/10.1177/0308518X16677969>.
- Lim, H., Kim, J., Potter, C., Bae, W., 2013. Urban regeneration and gentrification: land use impacts of the Cheonggye Stream Restoration Project on the Seoul's central business district. *Habitat Int.* 39, 192–200. <https://doi.org/10.1016/j.habitatint.2012.12.004>.
- Lin, G.C., 2002. The growth and structural change of Chinese cities: a contextual and geographic analysis. *Cities* 19 (5), 299–316. [https://doi.org/10.1016/S0264-2751\(02\)00039-2](https://doi.org/10.1016/S0264-2751(02)00039-2).
- Liu, D., Richards, K., 2019. The He-Zhang (River chief/keeper) system: an innovation in China's water governance and management. *Int. J. River Basin Manag.* 17 (2), 263–270. <https://doi.org/10.1080/15715124.2018.1546716>.
- Logan, J.R., Molotch, H., 1987. *Urban fortunes: The political economy of place*. University of California Press, Berkeley, CA.
- Long, F., Liu, J., Zheng, L., 2022. The effects of public environmental concern on urban-rural environmental inequality: evidence from Chinese industrial enterprises. *Sust. Cities Soc.* 80, 103787. <https://doi.org/10.1016/j.scs.2022.103787>.
- Long, H., Liu, Y., 2016. Rural restructuring in China. *J. Rural Stud.* 47, 387–391. <https://doi.org/10.1016/j.jrurstud.2016.07.028>.
- MacLeod, G., 2011. Urban politics reconsidered: growth machine to post-democratic city? *Urban Stud.* 48 (12), 2629–2660. <https://doi.org/10.1177/0042098011415715>.
- MacLeod, G., Goodwin, M., 1999. Reconstructing an urban and regional political economy: on the state, politics, scale, and explanation. *Polit. Geogr.* 18 (6), 697–730. [https://doi.org/10.1016/S0962-6298\(99\)00014-1](https://doi.org/10.1016/S0962-6298(99)00014-1).
- Molotch, H., 1976. The city as a growth machine: toward a political economy of place. *Am. J. Sociol.* 82 (2), 309–332. <http://www.jstor.org/stable/2777096>.
- Molotch, H., 1993. The political economy of growth machines. *J. Urban Aff.* 15 (1), 29–53. <https://doi.org/10.1111/j.1467-9906.1993.tb00301.x>.
- Munshi, D., Kurian, P., 2005. Imperializing spin cycles: a postcolonial look at public relations, greenwashing, and the separation of publics. *Public Relat. Rev.* 31 (4), 513–520. <https://doi.org/10.1016/j.pubrev.2005.08.010>.
- Oi, J.C., 1995. The role of the local state in China's transitional economy. *China Q.* 144, 1132–1149. <https://doi.org/10.1017/S030574100004768>.
- Ostrom, E., 2010a. Beyond markets and states: polycentric governance of complex economic systems. *Am. Econ. Rev.* 100 (3), 641–672. <https://doi.org/10.1257/aer.100.3.641>.
- Ostrom, E., Gardner, R., 1993. Coping with asymmetries in the commons: self-governing irrigation systems can work. *J. Econ. Perspect.* 7 (4), 93–112. <https://doi.org/10.1257/jep.7.4.93>.
- Ostrom, E., 2010b. Polycentric systems for coping with collective action and global environmental change. *Glob. Environ. Change, Hum. Pol. Dimens.* 20 (4), 550–557. <https://doi.org/10.1016/j.gloenvcha.2010.07.004>.
- Qi, Y., Zhang, L., 2014. Local environmental enforcement constrained by central-local relations in China. *Environ. Policy Gov.* 24 (3), 216–232. <https://doi.org/10.1002/eet.1640>.
- Rigolon, A., Németh, J., 2018. “We're not in the business of housing:” environmental gentrification and the nonproliferation of green infrastructure projects. *Cities* 81, 71–80. <https://doi.org/10.1016/j.cities.2018.03.016>.
- Sbicca, J., 2019. Urban agriculture, revalorization, and green gentrification in Denver, Colorado. *Politics Land* 26, 149–170. <https://doi.org/10.1108/S0895-99352019000026011>.
- Scanu, E., Cloutier, G., Trudelle, C., 2021. Contesting the greening of the urban growth machine: ecological modernization and the promethean counter-discourse. *Environ. Plan. E Nat. Space* 4 (4), 1369–1388. <https://doi.org/10.1177/2514848620952326>.
- Scott, M., 2004. Building institutional capacity in rural Northern Ireland: the role of partnership governance in the LEADER II programme. *J. Rural Stud.* 20 (1), 49–59. [https://doi.org/10.1016/S0743-0167\(03\)00042-1](https://doi.org/10.1016/S0743-0167(03)00042-1).
- Shao, Z., Zhang, Y., Li, Y., 2025. Greening China's urban growth machine: the micro-politics of growth and environment protection in Wuxi, China. *J. Urban Aff.* 47 (3), 760–774. <https://doi.org/10.1080/07352166.2023.2208756>.
- Shen, M., Shen, J., 2018. Evaluating the cooperative and family farm programs in China: a rural governance perspective. *Land Use Pol.* 79, 240–250. <https://doi.org/10.1016/j.landusepol.2018.08.006>.
- Triguero-Mas, M., Angelovski, I., García-Lamarca, M., Argüelles, L., Perez-del-Pulgar, C., Shokry, G., et al., 2021. Natural outdoor environments' health effects in gentrifying neighborhoods: disruptive green landscapes for underprivileged neighborhood residents. *Soc. Sci. Med.* 279, 113964. <https://doi.org/10.1016/j.socscimed.2021.113964>.
- Tu, C., Liang, Y., Fu, Y., 2024. How does the environmental attention of local governments affect regional green development? Empirical evidence from local governments in China. *Hum. Soc. Sci. Commun.* 11 (1), 1–14. <https://doi.org/10.1057/s41599-024-02887-9>.
- Wang, C., Engels, A., Wang, Z., 2018. Overview of research on China's transition to low-carbon development: the role of cities, technologies, industries and the energy system. *Renew. Sust. Energ. Rev.* 81, 1350–1364. <https://doi.org/10.1016/j.rser.2017.05.099>.
- Wang, C.M., Maye, D., Woods, M., 2023. Planetary rural geographies. *Dialogues Hum. Geogr.* 20438206231191731. <https://doi.org/10.1177/20438206231191731>.
- Wang, K., Qian, J., Chen, C., Zhang, H., 2025. Variegated forms of planetary ruralisation: a multiscale contextual analysis of evolving ruralities of China's Dike-Pond System. *J. Rural Stud.* 119, 103786. <https://doi.org/10.1016/j.jrurstud.2025.103786>.
- Wang, M., Webber, M., Finlayson, B., Barnett, J., 2008. Rural industries and water pollution in China. *J. Environ. Manage.* 86 (4), 648–659. <https://doi.org/10.1016/j.jenvman.2006.12.019>.
- Wang, Y., Chen, X., 2020. River chief system as a collaborative water governance approach in China. *Int. J. Water Resour. Dev.* 36 (4), 610–630. <https://doi.org/10.1080/07900627.2019.1680351>.
- While, A., Jonas, A.E., Gibbs, D., 2004. The environment and the entrepreneurial city: searching for the urban ‘sustainability fix’ in Manchester and Leeds. *Int. J. Urban Reg. Res.* 28 (3), 549–569. <https://doi.org/10.1111/j.0309-1317.2004.00535.x>.
- Wu, J., Xiang, W., Zhao, J., 2014. Urban ecology in China: historical developments and future directions. *Landsc. Urban Plan.* 125, 222–233. <https://doi.org/10.1016/j.landurbplan.2014.02.010>.
- Xu, C., 2011. The fundamental institutions of China's reforms and development. *J. Econ. Lit.* 49 (4), 1076–1151. <https://doi.org/10.1257/jel.49.4.1076>.

- Xue, B., Han, B., Li, H., Gou, X., Yang, H., Thomas, H., Stücker, S., 2023. Understanding ecological civilization in China: from political context to science. *Ambio* 52, 1895–1909. <https://doi.org/10.1007/s13280-023-01897-2>.
- Yan, S., Chen, C., 2024. Rural transformation and sustainable livelihood approaches under regional environmental policies: evidence from two villages in the Yangtze River Delta metropolitan region, China. *Sustain. Dev.* 33 (2), 2096–2117. <https://doi.org/10.1002/sd.3195>.
- Ye, C., Schröder, P., Yang, D., Chen, M., Cui, C., Zhuang, L., 2022. Toward healthy and liveable cities: a new framework linking public health to urbanization. *Environ. Res. Lett.* 17 (6), 064035. <https://doi.org/10.1088/1748-9326/ac70eb>.
- Ye, C., Liu, Z., 2020. Rural-urban co-governance: Multi-scale practice. *Sci. Bull.* 65 (10), 778–780. <https://doi.org/10.1016/j.scib.2020.02.021>.
- Yu, J., Xian, Q., Cheng, S., Chen, J., 2024. Horizontal ecological compensation policy and water pollution governance: evidence from cross-border cooperation in China. *Environ. Impact Assess. Rev.* 105, 107367. <https://doi.org/10.1016/j.eiar.2023.107367>.
- Yu, Y., Li, K., Duan, S., Song, C., 2023. Economic growth and environmental pollution in China: new evidence from government work reports. *Energy Econ.* 124, 106803. <https://doi.org/10.1016/j.eneco.2023.106803>.
- Yu, J., Ye, C., 2025. A new conceptual framework and case analysis of rural environmental cross-boundary governance in megacities of China. *Habitat Int.* 156, 103258. <https://doi.org/10.1016/j.habitatint.2024.103258>.
- Zhang, S., 2014. Land-centered urban politics in transitional China—Can they be explained by Growth Machine Theory? *Cities* 41, 179–186. <https://doi.org/10.1016/j.cities.2014.02.010>.
- Zhang, X., 2017. Implementation of pollution control targets in China: has a centralized enforcement approach worked? *China Q.* 231, 749–774. <https://doi.org/10.1017/S0305741017000959>.
- Zhang, C., Tao, R., Yue, Z., Su, F., 2023a. Regional competition, rural pollution haven and environmental injustice in China. *Ecol. Econ.* 204, 107669. <https://doi.org/10.1016/j.ecolecon.2022.107669>.
- Zhang, F., Wu, F., Lin, Y., 2022a. The socio-ecological fix by multi-scalar states: the development of ‘Greenways of Paradise’ in Chengdu. *Polit. Geogr.* 98, 102736. <https://doi.org/10.1016/j.polgeo.2022.102736>.
- Zhang, F., Wu, F., Liu, Y., 2023b. China's urban environmental governance. In: Zhang, F., Wu, F. (Eds.), *Handbook on China's Urban Environmental Governance*. Edward Elgar Publishing, UK.
- Zhang, Q., Ye, C., Duan, J., 2022b. Multi-dimensional superposition: rural collaborative governance in Liushe village, Suzhou city. *J. Rural Stud.* 96, 141–153. <https://doi.org/10.1016/j.jrurstud.2022.10.002>.
- Zhang, J.X., Gao, Y., 2022. From growth coalition to development coalition: a case study of urban renewal governance transformation in Southern Area of Nanjing (SAN), China. *Trans. Plan. Urban Res.* 1 (1–2), 66–85. <https://doi.org/10.1177/27541223221115930>.
- Zhang, F., Wu, F., 2024. Green state entrepreneurialism: building the park city in Chengdu, China. *Trans. Plan. Urban Res.* 3 (3), 183–201. <https://doi.org/10.1177/27541223241253231>.
- Zhao, Y., 2024. Tourism dreams in rubble: mass demolition and the reconfiguration of growth coalitions within China's ecological civilization. *Environ. Plan. E Nat. Space* 7 (4), 1672–1690. <https://doi.org/10.1177/25148486241246942>.
- Zhao, X., Sun, H., Chen, B., Xia, X., Li, P., 2019. China's rural human settlements: qualitative evaluation, quantitative analysis and policy implications. *Ecol. Indic.* 105, 398–405. <https://doi.org/10.1016/j.ecolind.2018.01.006>.
- Zhou, L., 2007. Governing China's local officials: an analysis of promotion tournament model. *Econ. Res. J.* 7 (36), 36–50 [in Chinese].
- Zhu, X., Qiu, T., Liu, D., 2025. Curbing elite capture or enhancing resources: recentralizing local environmental enforcement in China. *China Q.* 261, 73–92. <https://doi.org/10.1017/S0305741024000730>.